

Accounting for Ocean Impacts Nearly Doubles the Social Cost of Carbon

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Abstract

Oceans provide essential benefits to people and the economy, underpinned by the extent and condition of marine ecosystems and infrastructure or “blue” capital. However, the impacts of climate change on blue capital have been largely overlooked in influential indicators such as the social cost of carbon (SCC). Here, we integrate the latest ocean science and economics into a climate-economy model, capturing climate change impacts on corals, mangroves, seaports, fisheries, and mariculture to estimate their welfare repercussions at a global scale. Conceptually, this ocean-based SCC (blue SCC) represents a component of the total SCC currently omitted in standard estimates. We estimate the blue SCC to be \$48/tCO₂ [38-70, 25th-75th percentile] with baseline discounting, and more than three times as high for an annual rate of 2%. This represents an almost doubling of the SCC estimate from the same model and the existing literature without considering ocean-related impacts.

Main Text

The ocean is key to a thriving society. It provides a range of benefits that support market economies and sustain livelihoods and wellbeing¹⁻⁴. Some economic benefits derived from the marine environment, such as seafood and maritime transportation, are traded in markets and their value is observable in market prices; whereas others are not traded in markets and their economic values are not directly observable, e.g., coastal protection, recreational activities, and the values that people place on the existence of marine biodiversity. In response, a growing body of academic and policy-oriented literature on the multifaceted concept of the Blue Economy highlights the need for comprehensive frameworks that reflect the critical interdependence of the oceans, people, and the economy^{1,2,5}.

Examples of “blue capital” include fish biomass, living coral coverage, mangrove coverage, and seaports (while considered a human-made capital, they enable the ocean-based benefit of navigable waters; Figure 1a). The scatterplot of Figure 1b reports country-level blue-capital benefits computed with Eqs. (2)–(3) (Online Methods) and divided by GDP. The magnitude and composition of blue capital is distributed across geographies and socioeconomic groups, but the importance of blue capital, as a share of GDP, to the economy tends to be highest in low income countries (larger markers in Fig 1b occurring at lower GDP per capita).

Anthropogenic greenhouse gas emissions are changing the ocean's chemistry and physical conditions. Elevated greenhouse gas concentrations in the atmosphere leads to higher ocean temperatures⁶, reduces the ocean's ability to hold oxygen⁷, causes ocean acidification⁸, increases the severity of extreme weather events⁹, and contributes to sea level rise. These changes to the ocean are driving widespread alterations to species distributions, phenology, body sizes, ecosystem structures and functions, and mass mortalities of organisms. Some of the critical habitats for biodiversity and ecosystem services are particularly vulnerable to climate change, resulting in large-scale bleaching of coral reefs and loss of seagrass beds and kelp forests. These changes in the ocean directly affect the benefits we derive from it and should be factored into tools and frameworks that support climate policy and regulatory efforts.

The social cost of carbon (SCC), a well-known metric in climate policy and regulatory analysis, is an estimate of the total economic impact on humanity from the release of an additional tonne of CO₂ into the atmosphere. It should encompass all known climate-related damages to people and the biosphere. While integrated assessment models (IAMs) to calculate the SCC typically include an ocean carbon module for the physical regulation of CO₂, the economic damages to marine ecosystems and ocean-dependent activities have been absent from calculations. Recent analyses and systematic reviews acknowledge the omission of known climate-related ocean damages and call for their integration^{10–13}. For instance, a systematic review by the U.S. National Academy of Sciences¹⁰ emphasized the need for a more accurate representation of climate-related damages, including impacts on marine fisheries, coral-based recreation, and tourism services, which are currently not represented in existing models. Similarly, the U.S. Environmental Protection Agency's 2023 technical report conducts a thorough review of the SCC calculation, pointing out the absence of any representation of marine ecosystem services and fisheries damages¹². Including ocean impacts in the SCC is essential for designing environmental policies that accurately reflect the economic value of greenhouse gas emissions. In this work, we present the first calculation of an ocean-based SCC (hereafter ‘blue SCC’).

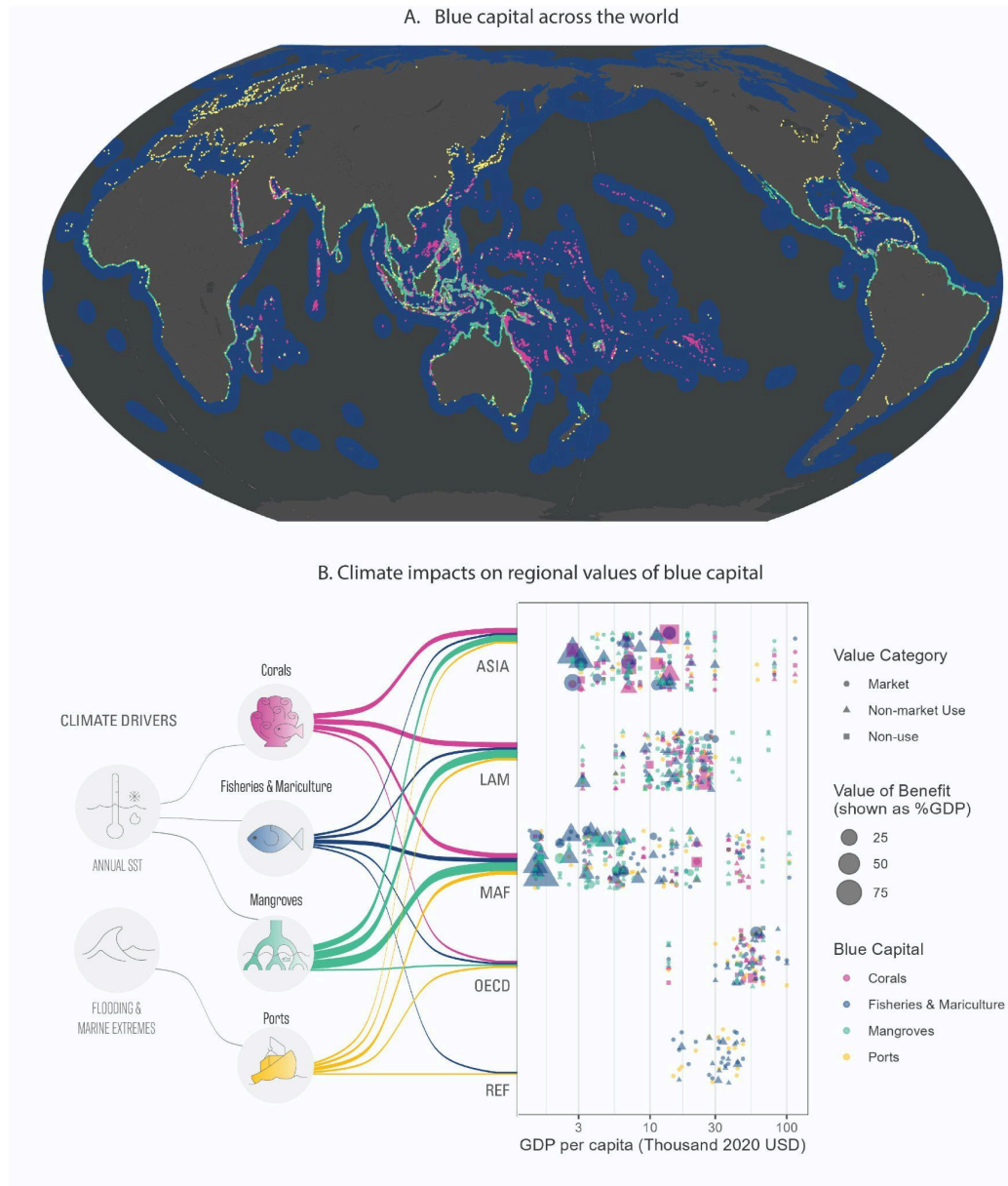


Figure 1. Blue capital and its values (in 2020, or most recent year for which data are available). Panel A shows the geographical distribution of the four types of blue capital considered in this study: corals, mangroves, fisheries and mariculture, and seaports. Panel B is a flow diagram that reads from left to right. The leftmost set of icons represents the climate drivers through which climate change acts on blue capital in this study. The gray lines indicate the impact pathways through which climate change affects blue capital, as identified in the literature, the colored lines show the distribution of blue capital across socioeconomic regions where the line width indicates the proportion of countries within each region that have such blue capital. The five regions are: ASIA, Asian countries except the Middle East, Japan and the former Soviet Union states; LAM, Latin America and the Caribbean; MAF, the Middle East and Africa; OECD, the OECD countries and the European Union member states and candidates; REF, the reforming economies of Eastern Europe and the former Soviet Union. In the rightmost scatterplot annual flow of blue capital values at the country-level, as computed by equations (1) and (2), are shown by country as a percentage of Gross Domestic Product (GDP), ordered by GDP per capita.

Approach

Our analysis integrates findings from the climate change impacts literature on corals, mangroves, seaports, and fisheries and mariculture— four sectors for which economic impact assessment studies have been conducted at a sufficiently detailed spatial scale to estimate country-level or subnational impacts. Our process harmonizes information from six main disparate studies into a framework that enables the quantification of the ocean-based economic damages associated with incremental warming, a necessary requirement to estimate the SCC (Figure 2). In these sectoral impact studies, climate variables such as sea surface temperatures (annual mean and during the warmest month), maritime extreme events, and coastal flooding are empirically linked to impacts on blue capital and statistically linked to global mean temperature.

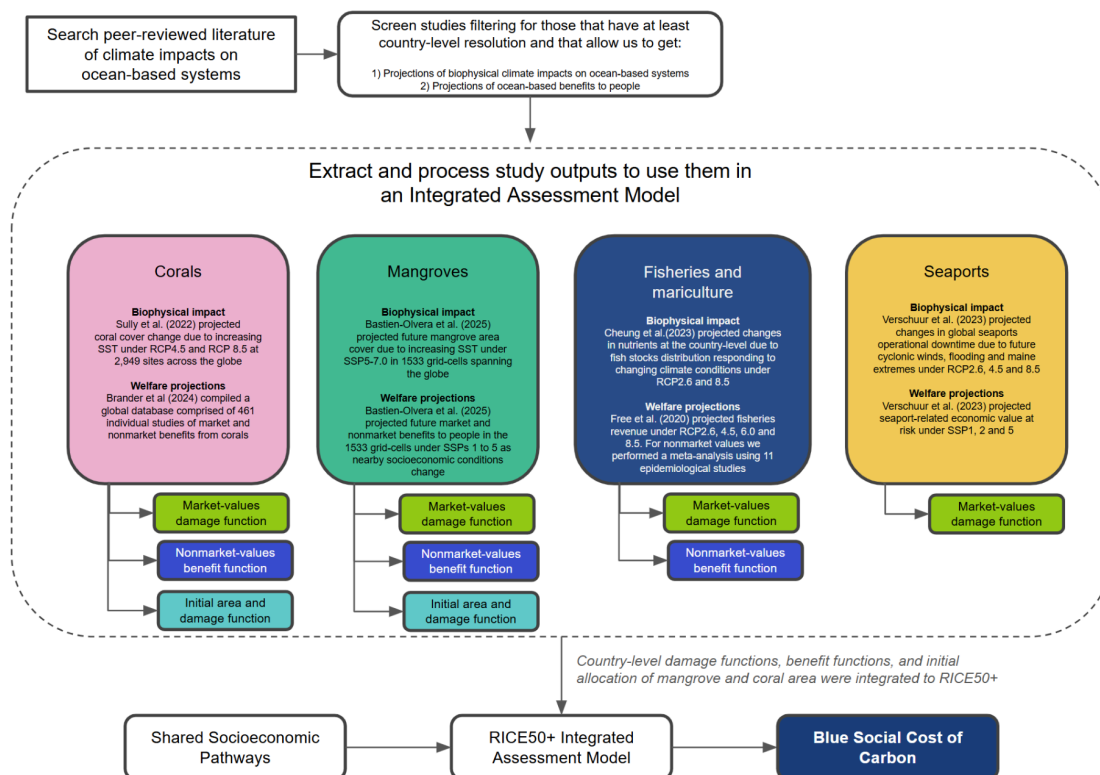


Figure 2. Analytical framework for estimating the ocean-based social cost of carbon.

This diagram presents the structured analytical framework developed to quantify the Blue SCC by integrating biophysical and economic valuation functions. The process starts with a literature search, screening studies that quantify climate-induced impacts on ocean-based assets or marine ecosystems. An in-depth analysis follows, focusing on the four types of blue capital identified: corals, mangroves, fisheries and mariculture, and seaports. Within each capital block, the diagram briefly presents the underlying biophysical and economic valuation studies that were selected. Each block highlights the types of values accounted for: market values, non-market use values, and non-use values, as applicable. The processed values from each block are aggregated at the country level and subsequently integrated into

the RICE50+ IAM. In the model, users can select the desired Shared Socioeconomic Pathway (SSP) to project the Blue Social Cost of Carbon under different climate and socioeconomic scenarios.

Our valuation of climate-induced damages to blue natural capital encompasses impacts on human well-being that are directly and indirectly reflected in markets. Market values are relatively straightforward, encompassing monetary losses such as decreased fisheries revenue or diminished trade. Non-market values include human health and nutrition from fisheries, coastal protection, recreational opportunities, as well as non-use values, generally termed existence value, that reflect the inherent worth that humans ascribe to ecosystems and biodiversity. We translate the projected changes in all blue capital categories under various climate change scenarios into welfare losses, and quantify these into monetary equivalents, allowing for a comprehensive analysis of ocean climate damages.

To account for market-related damages within our framework, we adopt a damage function approach commonly used in IAMs. This method uses stylized functions that relate the projections of country-level market loss to increases in global mean surface temperature relative to pre-industrial levels. Non-market values are incorporated from a natural capital perspective, meaning the climate-induced impacts on the stock of blue natural capital in each country (measured as mangroves and coral area coverage) are assigned a monetary value equivalent to the welfare benefits it provides. We call *benefit function* the relationship that maps ecosystem stock into a flow of benefits. These two pathways—the damage function and the natural capital approach—were integrated into RICE50+¹⁴, a computational model that simulates the dynamics of countries' economies under the influence of climate change. This integration resulted in a new version of the model, referred to as the “Blue” RICE50+ model, which specifically incorporates blue capital and is capable of calculating the SCC for the oceans. Details can be found in the Methods and Supplementary Information.

The economic damages of climate change to the oceans

We present the impacts to ocean-based welfare under a scenario representing a middle-of-the-road socioeconomic development path (SSP2) that, under a no-policy scenario, leads to an average temperature increase of approximately 3.6°C by the end of the century over preindustrial levels. Emissions and temperatures are determined endogenously by RICE50+, rather than by pairing SSP2 with an exogenous forcing pathway (e.g., RCP4.5). We explore alternative socio-economic and climate scenarios in the Appendix (Figure S24).

The annual country-level damage trajectories from 2020 to 2100 (Fig. 3a) vary across market use, non-market use, and non-use values. Damages increase over time as economies grow. Overall, market use damages are the largest in absolute terms, totaling global annual losses of US\$1.66 trillion in 2100, followed by damages in non-use values amounting to US\$224 and non-market use values adding up to \$182 billion in annual losses.

Non-market damages are primarily driven by the health impacts of reduced nutrition availability from fisheries and mariculture, access to which substantially impacts mortality and morbidity, especially in lower-income countries. Non-market use value losses from corals and mangroves, which provide important coastal protection services, are also large. The largest share of damages in non-use values comes from the loss of mangroves in India, where the impact on mangroves is significant and the local population benefiting from these non-use values is substantial and wealthy.

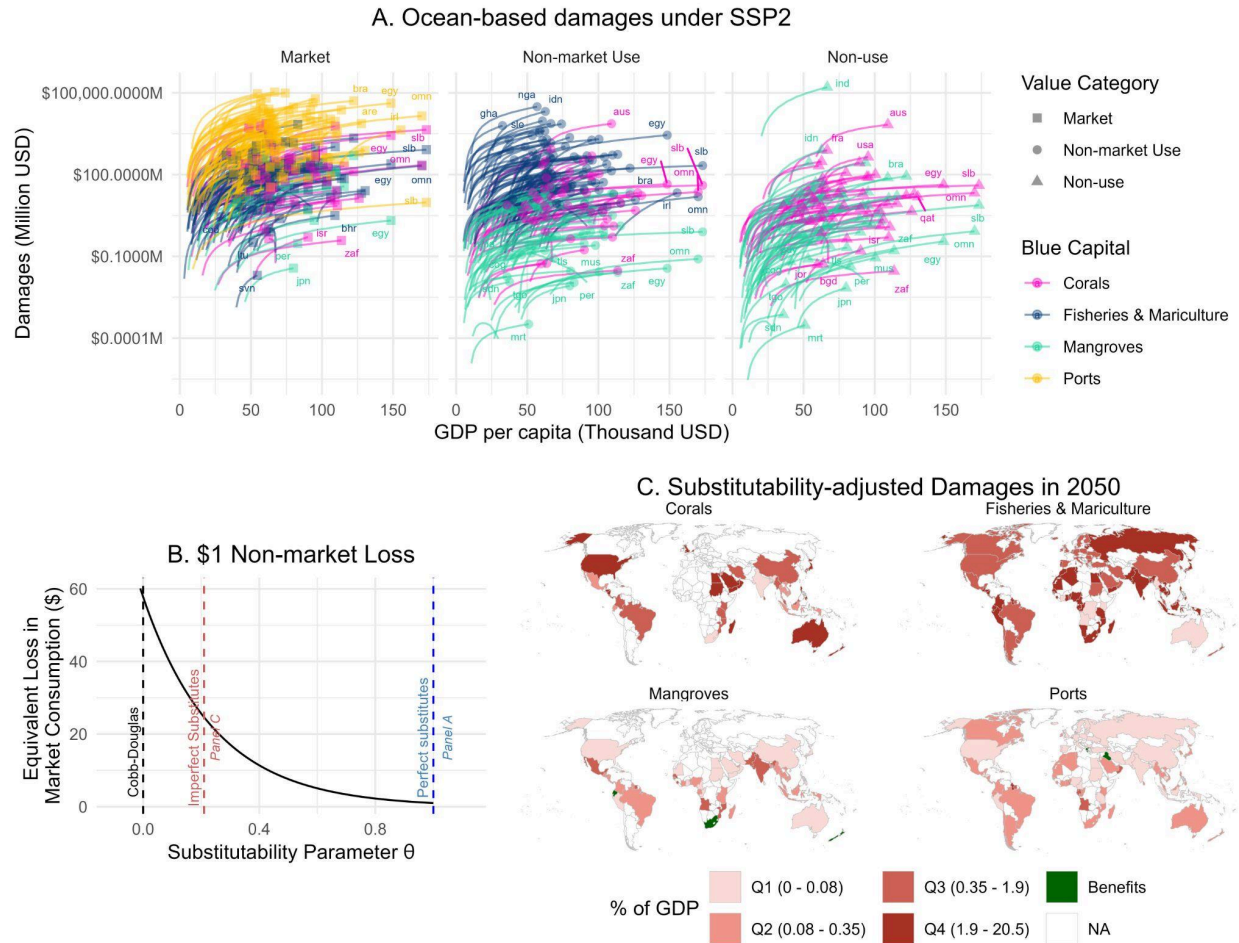


Figure 3. Climate damages on ocean-based values.

This figure presents the projected economic damages resulting from climate change impacts on blue capital, broken down into three panels. (A) Country-level damage trajectories in billions of US\$ (log-scale) for each country across different types of values, from 2020 to 2100. Each country's path is a plain line starting at 2020 (unmarked end) and ending at 2100, which is indicated by a terminal marker — square for Market, circle for Non-market use, triangle for Non-use. (B) The effect of the substitutability parameter on the market consumption-equivalent of losing \$1 of non-market values. The values shown are given by using Australia's market consumption levels and the nonmarket coral benefits in 2025 as an example.

Theta can also be negative, representing various levels of complementarity; as theta approaches negative infinity, the value of one non-market dollar tends toward infinity, corresponding to the case of perfect complements. This illustrative example is not tied to a specific nesting level within the utility function; it serves to build intuition about how substitutability affects welfare valuation (See SM for further

intuition). (C) Total climate damages in 2050 aggregated across the four types of blue capital shown in quantile groups relative to national GDPs.

One unit of loss in market-equivalent value is not equivalent to one unit of loss in non-market welfare, because market and nonmarket values are not perfect substitutes (Fig. 3B). In terms of the substitutability parameter, we use a conservative elasticity of $\theta = 0.21$ based on recent literature¹⁵. This is a very important parameter as shown in the uncertainty analysis.

The relative magnitudes of substitutability-adjusted damages are substantially different from those in absolute terms (Fig. 3C for the year 2050, see also Fig S27). Each type of capital: corals, seaports, fisheries and mariculture, and mangroves is associated with market, non-market use, and non-use damages, where applicable. Ports have only market damages, whereas other capitals (such as mangroves and corals) include all the three value types (See Table 1). The highest welfare-adjusted damages are for fisheries and mariculture, while welfare-adjusted damages associated with ports are generally the lowest. A few countries have expected net benefits.

The blue Social Cost of Carbon

To calculate the blue SCC, we first run the model under a reference scenario with climate impacts and then a second time with an additional megaton pulse of CO₂ in 2020. We aggregate the different impacts and combine the economic and welfare effects through a utility function with decreasing marginal utility with income. We normalize the results per ton of CO₂, computing the net present value of future impact streams using an endogenous discount factor (see Methods). We keep the baseline model parameters as central specification, and explore different values as sensitivity (see below and SI).

We find that the 2020 blue SCC, calculated as non equity-weighted global average, is \$48 per ton of CO₂ (Fig. 4a). For a pulse of CO₂ in 2030, the blue SCC is \$72 (all values are expressed in 2020 U.S. dollars). Roughly half of the blue SCC can be attributed to non-market-related losses from fisheries and mariculture (namely their health costs), with the other half stemming from corals and mangroves. Notably, the welfare impacts from changes to the market use values of ports are small. This does not imply that the accumulated damages from these sources are insignificant, but rather reflects the essential welfare role of oceans' non-market use and non-use services given their limited substitutability (Fig. 3) as well as their typology and geographical distribution. For example, health impacts are the largest component of the blue SCC because (i) they have a high economic value (see Methods) and (ii) the reliance on essential nutrients from the oceans is concentrated in low-income countries, which have a high marginal utility.

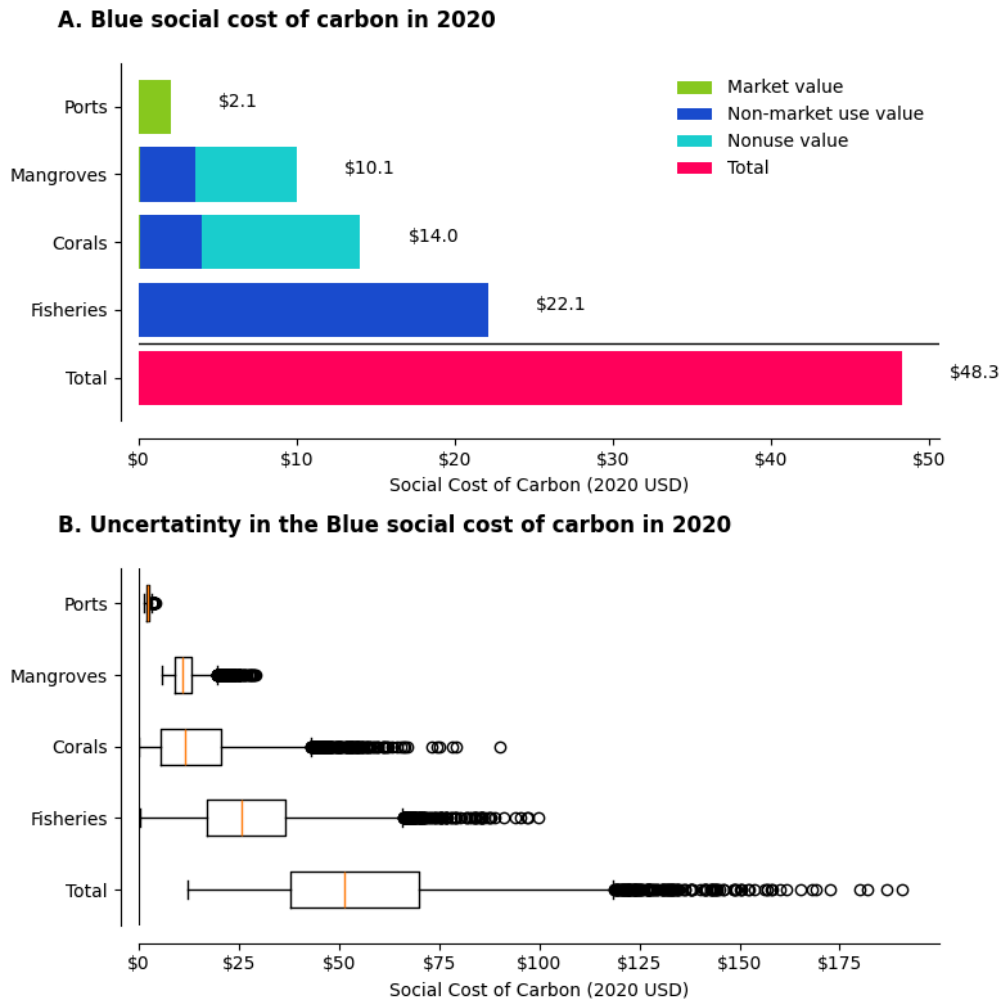


Figure 4. Blue Social Cost of Carbon and its uncertainty.

This figure presents the breakdown of the Blue Social Cost of Carbon (SCC) and the sensitivity of the results to input parameters.

Top Panel: The Blue SCC is disaggregated by sector (rows), with each row corresponding to a different sector of blue capital (corals, mangroves, seaports, fisheries, and mariculture), while each color within the rows represents a different value type (market use, non-market use, non-use). The total SCC is displayed at the bottom as an aggregate of all components.

Bottom Panel: The sensitivity analysis shows the distribution of Blue SCC estimates resulting from 10,000 simulations using a Monte Carlo approach. Boxes span the interquartile range (from the first quartile (Q1) to the third quartile (Q3)) with the orange line representing the median. The mean value of \$48/tCO₂ not shown in the graph. Whiskers extend to the farthest data points within 1.5 times the interquartile range from the box, while circles indicate outliers. Values of the blue SCC for different pure rates of time preference are shown in Extended Data Figure 1.

To quantify uncertainty, we use a Monte Carlo approach to calculate blue SCC with 10,000 combinations of parameter values (See Figure S19). This allows us to quantify and partition uncertainty associated with the damage functions, ecosystem services value functions, the degree of substitutability between market, non-market use, and non-use values, as well as

climate sensitivity, the pure rate of time preference, and the elasticity of the marginal utility of consumption. Overall, the interquartile range (25th to 75th percentile) of the blue SCC is estimated between \$38 and \$70 USD per ton of CO₂. This suggests that, despite the inherent uncertainties, the SCC for ocean-based welfare represents a magnitude comparable—if taken at face value—with other large sector-specific SCC estimates (e.g., \$84 for agriculture, \$9 for energy, and \$90 for mortality as given by the GIVE model¹⁶ or \$4 for agriculture, -\$3 for energy, \$39 for labor, \$3 for coastal damages due to sea level rise and \$146 for mortality as given by the DSCIM model^{17–19}, both used by the 2023 EPA report¹¹). The most important driver of uncertainty in the blue SCC is the elasticity of substitution between market and non-market values in the utility function (see Figure S19). The SCC is also very sensitive to parameters governing the non-use value of corals and the non-market use value of fisheries. We also compute the blue SCC across different SSPs, finding a skewed range between \$37 (SSP5) and \$77 (SSP3) USD per ton of CO₂ (Fig S25).

Finally, we recalibrate the pure rate of time preference and elasticity of marginal utility of consumption to be consistent with near-term consumption discount rates of 1.5%, 2%, 2.5%, and 3%, ensuring comparability with more recent work.¹⁶ The resulting estimates are \$305, \$168, \$96, and \$64 USD₂₀₂₀/tCO₂ for the values above. For a rate of 2%/year, the consensus value among experts²⁰, the blue SCC is more than three times the baseline value and is comparable to the SCC without ocean impacts.

Discussion and conclusions

There has been a long-standing need to integrate marine ecological responses to climate change into economic metrics that inform policymaking. While existing IAMs typically simulate ocean carbon uptake dynamics, they do not explicitly model damages to ocean ecosystems or ocean-based sectors. Some work has explored climate damages linked to ocean dynamics (e.g., AMOC slowdown^{21,22}). However, this is the first estimate that comprehensively integrates climate damages to a range of ocean-based ecological and economic systems into the SCC framework. Moreover, while recent econometric work implicitly captures climate impacts on market sectors in regional GDP projections, our approach disentangles sector-specific pathways by explicitly modeling market damages from blue capital degradation. The distribution of ocean-based welfare impacts is highly unequal across the world, with islands and small economies being disproportionately affected, a pattern also observed for terrestrial-based ecosystems²³.

The blue SCC presented here implies a substantial increase relative to the SCC obtained when running the standard RICE50+ model without ocean-based damages. Specifically, our results indicate that including ocean-based welfare would almost double the standard SCC derived from the same integrated assessment model (\$51 per ton of CO₂, in the central case, see SM). When we calibrate our model to the near-term 2% discount rate used in recent SCC estimates

from the EPA (applied in U.S. policy as of 2024¹¹) our result (\$168) represents an 88% increase in the EPA's SCC. However, this comparison to EPA estimates should be interpreted with caution due to key methodological inconsistencies (see SM). This increase is also consistent with trends in SCC estimates, which have historically risen as new scientific knowledge is incorporated²⁴. It also responds to numerous calls to update the scientific foundations underlying current SCC estimates^{25,26}, many of which explicitly call for more comprehensive ocean-based assessments^{10,11,13,23}. Our estimate expands on the existing estimates by incorporating several ocean-based damages that the EPA acknowledges as important but has not yet included in SCC calculations, such as impacts on fisheries and mariculture, tourism, recreation, and aesthetic values. The latest SCC report from the U.S. National Academy of Science recommended the inclusion of an ocean chemistry module, a step toward accounting for ocean-based damages driven by indicators that are not closely coupled with temperature. Our analytical framework can be expanded to meet this goal. Blue SCC should be conceptually understood as a component of the total SCC, which includes all climate damages. However, because most current estimates omit ocean-based damages entirely, our results can be interpreted as the degree to which existing SCC estimates may be underestimating the true social cost of carbon emissions.

Diverse climate drivers underlie the biophysical impacts on each type of blue capital analyzed in this study. While we standardize these impacts by re-scaling them to global mean surface temperature for ease of implementation in IAMs, including RICE50+, the underlying science is far more complex. For coral reefs, the estimates are driven by localized sea surface temperature changes²⁷; for mangroves, the key driver is the sea surface temperature of the hottest month²⁸. Seaport infrastructure is influenced by a combination of pluvial, riverine and coastal flooding as well as cyclone and other extreme weather events²⁹. In the case of fisheries and mariculture, our approach relies on a process-based model of marine ecosystems—specifically, the dynamic bioclimate envelope model—which incorporates sea surface temperature, sea bottom temperature, coastal upwelling, salinity, and other environmental factors^{30,31}. Future implementations can retain these sector-specific local drivers and link them to IAMs via spatial patterns; the GMST rescaling used here serves as a common baseline.

Though based on the best global knowledge, our analysis could be further refined to disentangle ocean drivers. For instance, sea surface temperature increases contribute to coral bleaching by disrupting the symbiotic relationship between corals and algae. Simultaneously, ocean acidification impacts them by reducing their ability to form and maintain their structures³². While these two drivers are often correlated in climate projections³³, disentangling their specific effects could be valuable for exploring alternative scenarios where mitigation policies might target one but not the other. For example, solar radiation management could reduce sea surface temperature without addressing acidification.

The role of ecosystem services in supporting society and the economy has recently gained recognition as fundamental for policy making¹⁵. Using the latest empirical evidence on substitution elasticities—which capture the limited potential to replace natural services with human-made alternatives³⁴—and employing welfare functions consistent with dual discounting^{35,36} and previous IAMs^{37,38}, we find that the non-market use and non-use values of

ocean-based ecosystems are, the largest determinants of welfare losses from climate change, as reflected in the blue SCC. This metric can also strengthen complementary efforts to assess ocean wealth. For example, Bertram et al. (2021)³⁹ estimate ocean value by applying existing SCCs to the carbon sequestration of coastal ecosystems; incorporating a blue SCC would capture both carbon-storage benefits and climate damages to marine ecosystems, yielding a more complete ocean valuation.

Future work should broaden blue capital coverage and benefit types; for instance, our fisheries values currently include only economic revenue and nutrition. However, for many communities, being "people of the sea" and practicing traditional fishing methods is central to their cultural identity—something that cannot be captured by the utility approach alone.^{40,41} Moreover, future estimates should incorporate additional types of blue capital, such as deep-sea ecosystems, which provide both use and non-use values.⁴² These contributions are largely unaccounted for in the SCC and, as a result, are often overlooked in regulatory analyses that determine climate outcomes.

As the ocean takes a central role in climate policy, from carbon sequestration to broader interventions,^{43,44} it is crucial to address both the impacts of policy actions and the direct effects of climate change on ocean ecosystems. Only by integrating these perspectives can we fully understand the challenges and opportunities of preserving, and where possible restoring, our blue capital—a critical step toward achieving sustainable ocean stewardship during the United Nations Decade of Ocean Science for Sustainable Development. With its potential for practical applications in regulatory analysis around the globe, an estimate of a blue SCC is one tool for synthesizing the global consequences detailed in the extensive literature on climate change impacts on the ocean. It also aids in designing environmental policies that account for the economic value of marine ecosystems.

Online Methods

Analytical Framework

To quantify the blue SCC, we developed an analytical framework designed to systematically integrate projections from biophysical and economic valuation studies available in the peer-reviewed literature. We conducted a structured literature search focused broadly on studies quantifying climate-induced impacts on ocean-based assets or marine ecosystems, with explicit consideration for those that provided corresponding economic welfare valuations—either directly, or through supplementary economic assessments or meta-analyses. Our search resulted in four sufficiently developed sectors—corals, mangroves, fisheries and mariculture, and seaports—that fulfilled two key criteria: (1) availability of geographically explicit biophysical projections at the national or subnational scale, and (2) associated economic valuations, enabling welfare impact estimation at comparable spatial resolutions.

Our integrative approach required matching two distinct methodological components: the biophysical impact functions and the economic valuation functions. The biophysical component captures climate-driven changes in asset availability (e.g., mangrove coverage, coral reef

health, fisheries productivity, port functionality), derived from peer-reviewed climate impact modeling studies. The economic component translates these biophysical changes into welfare impacts using valuation methods appropriate for different categories of value: market use, non-market use, and non-use. For studies already encompassing integrated biophysical and economic results, such as fisheries revenue impacts due to climate change or economic value-at-risk resulting from increased seaport downtime, we directly processed these datasets to generate reduced-form damage functions linking global mean surface temperature changes to country-level economic welfare impacts. For asset-value combinations requiring integration from multiple studies, we first constructed biophysical damage functions linking global temperature to physical impacts such as coral and mangrove area reductions. Subsequently, we developed simplified benefit functions directly relating these biophysical changes (e.g., ecosystem area) to welfare benefits, adjusting for projected increases in income.

The resulting damage functions facilitate integration into the modified Integrated Assessment Model RICE50+. This compatibility enables consistent and flexible scenario analysis under various Shared Socioeconomic Pathways (SSPs), supporting robust exploration of future climate-economic outcomes.

An essential feature of our analytical framework is its explicit consideration of the multiple dimensions of welfare impacts from ocean-based climate damages. We distinguish among three value types:

- Market values, where welfare impacts correspond directly to observable market prices, such as fishery revenues and maritime trade through ports.
- Non-market use values, representing benefits from direct ecosystem service use without explicit market pricing, such as nutritional health derived from fisheries or coastal protection from mangroves.
- Non-use values, reflecting welfare impacts stemming from the intrinsic existence of ecosystems independent of direct human use, such as biodiversity preservation in coral reefs.

Not all ocean-based assets exhibit all value categories due to either theoretical constraints or data limitations. For instance, non-use values are conceptually irrelevant for seaports, while non-use values for fisheries, although theoretically plausible, lack sufficient empirical quantification at the required spatial resolution. A detailed overview on how the valuation categories were implemented in RICE50+ for each sector is provided in Table 1, clearly delineating the mechanism of inclusion.

Table 1. Inclusion mechanisms for blue capital values in RICE50+

	Market values	Non-market use values	Non-use values
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Corals	Damage function linking global mean surface temperature with decrease in market consumption.	Damage function links global mean surface temperature with decrease in mangrove area Meta-regression using an ecosystem services database links area of ecosystem to value of nonmarket use benefits	Damage function links global mean surface temperature with decrease in mangrove area Meta-regression using an ecosystem services database links area of ecosystem to value of nonuse values
Mangroves	Damage function linking global mean surface temperature with decrease in market consumption.	Damage function links global mean surface temperature with decrease in mangrove area Meta-regression using an ecosystem services database links area of ecosystem to value of nonmarket use benefits	Damage function links global mean surface temperature with decrease in mangrove area Meta-regression using an ecosystem services database links area of ecosystem to value of nonuse values
Fisheries and Mariculture	Damage function linking global mean surface temperature with decrease in market consumption.	Nonmarket-value benefit function represents the change in nutritional value from fisheries as global mean surface temperature increases.	Insufficient data
Ports	Damage function linking global mean surface temperature with decrease in market consumption.	Insufficient data	No theoretical basis

RICE50+ Model structure

To calculate the social cost of carbon for ocean-based welfare, we use an integrated assessment model that links the increase of a marginal ton of carbon dioxide emissions to a series of interconnected systems. The approach begins with a carbon cycle model that describes the atmospheric response to additional emissions. The resulting changes are fed into a climate model (an implementation of the FaIR model⁴⁵), which simulates the corresponding temperature effects. Finally, these climatic changes are assessed for their impacts on ocean-based welfare, capturing the socioeconomic consequences of alterations in corals, mangroves, fisheries and ports.

For this analysis, we utilize the RICE50+ model (v2.6.0)¹⁴, an open-source coupled climate-economy model that extends the well-known Regional Integrated Climate-Economy (RICE) model³⁶. RICE50+ represents a dynamic framework that divides the world into more than 50 regions (57 in its standard version, here we run it at country level with 154 countries), allowing for region-specific assessments of the

costs and benefits of carbon emissions. The impact of climate change on natural capital from changes on terrestrial vegetation cover has been recently quantified with this model²³. We have enhanced this model to include the latest ecological and economic findings on the impacts of climate change on blue capital—the natural and human-made assets related to the ocean, as shown in Fig S1.

We modified the utility function (which, in the standard version, depends only on the level of consumption $C_{c,t}$ in country c at year t) to account for use and nonuse values of four different ocean-based systems: corals, mangroves, seaports, and fisheries and mariculture. We modified the utility function to take the form:

$$Utility_{c,t} = \frac{1}{1-\eta} \left[(1 - s_{2,1}) \cdot Use_{c,t}^{\theta_2} + s_{2,2} \cdot Nonuse_{c,t}^{\theta_2} \right]^{\frac{1-\eta}{\theta_2}} \quad \text{Eq. 1}$$

Where θ_2 is the substitutability parameter between use- and nonuse values^{34,47}. Note that in the RICE50+ framework, spatial inequality aversion can be separated from intertemporal preferences. Since here we don't implement global optimization scenarios, inequality aversion does not matter and we also don't consider equity weighting⁴⁸ since we are interested in the global uniform SCC. Therefore, we aggregate population-weighted country utilities converted into monetary terms and apply discounting as described in the section below. The nonuse value in Equation (1) is given by:

$$Nonuse_{c,t} = \sum_{b=mangroves, coral} V_{c,t}^{b, nonuse} \quad \text{Eq. 2}$$

where V represents the annual flow of non-use value associated with blue capital b (e.g., mangroves or corals) in each country.. Although non-use values in principle could be associated with ecosystems across countries, in our approach we assign each country's non-use value based solely on the area of ecosystems within its own territory and a corresponding shadow price, without transferring ecosystem values across countries. We further divided the use values into those that are exchanged in the market economy and those that are not, also acknowledging that these use values are imperfect substitutes, captured by the elasticity of substitution θ_1 :

$$Use_{c,t} = \left[(1 - s_{1,1}) \cdot C_{c,t}^{\theta_1} + s_{1,2} \cdot \left(\sum_{b=mangroves, coral, fisheries} V_{c,t}^{b, nonmarket use} \right)^{\theta_1} \right]^{\frac{1}{\theta_1}} \quad \text{Eq. 3}$$

Following the structure introduced by Hoel and Sterner⁴⁹, we retained the share parameters $s_{1,1}$, $s_{2,1}$, $s_{1,2}$ and $s_{2,2}$ in the utility function for consistency, but set $s_{1,1}$ and $s_{2,1}$ to zero, $s_{1,2}$ and $s_{2,2}$ to 1, as we explicitly calibrate the non-market components using empirical valuation data rather than rescaling

through arbitrary shares. The elasticities of substitution θ_1 and θ_2 is set to 0.21. The consumption C of all market goods and services is given by:

$$C_{c,t} = \tilde{C}_{c,t} \cdot \left[\left(1 + \lambda_{1,c}^{coral} + \lambda_{1,c}^{mangroves} + \lambda_{1,c}^{fisheries} + \lambda_{1,c}^{ports} \right) \cdot \Delta T_t + \left(1 + \lambda_{2,c}^{mangroves} \right) \cdot \Delta T_t^2 \right] \quad \text{Eq. 4}$$

Where $\tilde{C}_{c,t}$ represents the consumption level in the absence of climate change, and the coefficients $\lambda_{n,c}^b$, $n \in \{1, 2\}$, capture the relationship between temperature and market damages resulting from goods and services provided by corals, mangroves, fisheries, and ports.

Model parameterization

The value estimates that go into the utility function come from different approaches. For corals and mangroves, we first simulated the change in country-level cover area $A_{c,t}^b$ and then multiplied it by their shadow price $p_{c,t}^{b,v}$, with $b = \{\text{corals}, \text{mangroves}\}$ and $v = \{\text{nonuse}, \text{nonmarket use}\}$. Coral area cover changes as a linear function of temperature, based on Sully et al.²⁷, and the mangrove cover changes as a quadratic function of temperature, based on Bastien-Olvera et al.²⁸. The shadow price is obtained from the updated ecosystem services valuation database (ESVD) as presented in Brander et al.⁵⁰. Following Drupp et al.¹⁵, the value of corals market use and nonuse values increases over time with the rate of growth of GDP per capita as:

$$p_{c,t+1}^{corals,v} = p_{c,t}^{corals,v} \left(1 + \gamma^v \cdot g_{c,t} \right) \quad \text{Eq. 5}$$

where γ^v is the income elasticity estimated from a regression model using ESVD studies for market use and nonuse values. For mangroves, $\log(p_{c,t}^{mangroves,v}) = a_c^v + b_c^v \log(GDPpc_{c,t})$, where the parameters a_c^v and b_c^v re-scale the high resolution findings from Bastien-Olvera et al.²⁸, to average country-level value of mangroves using country-level GDP per capita. Therefore, the impact of climate change in these non-market values is only through reduced area, which would remain constant at today's values if temperature would not change. While in theory the marginal value of ecosystems may also rise as they become scarcer, we do not incorporate this effect here due to data limitations.

The market damage parameters $\lambda_{n,c}^b$ for corals and mangroves were estimated by running two sets of simulations. First, we calculated the GDP contribution of corals and mangroves by multiplying their shadow price $p_{c,t}^{b,v}$ by the area, and assuming no climate-induced damages to their areas. Then, we calculated the contribution under a scenario with climate damages. The difference between these two scenarios, as percentage of GDP, was fitted to the global mean surface temperature using a linear function for corals, and a quadratic function for mangroves, following the functional forms in their respective studies of origin. This λ represents a real market output loss used in the IAM's damage

function — not a theoretical correction to GDP. While our method uses shadow prices and is conceptually related to wealth accounting frameworks, we do not compute net present values or deduct capital loss from national income. Instead, we keep ecosystem stocks in physical terms over time and compute their evolving contribution to market, non-market, and non-use values each year.

The nonmarket use values from fisheries and mariculture were derived from Cheung et al.³¹ analysis of the impacts of climate change on nutrient availability from fisheries and mariculture at the country level. We combined these projections with epidemiological studies on the dose-response relationships between several health outcomes and the intake of omega-3, calcium, iron, and protein—the same nutrients analyzed by Cheung et al.³¹ With that, we projected changes in mortality rates for each country based on climate impacts on these nutrients, considering each country's level of dependence on fisheries, while allowing for a conservative assumption that a substantial portion of fish-derived nutrients can be replaced by other sources. Our preferred specification is to monetize these impacts using a globally uniform value of a statistical life (VSL). This is derived from the U.S. EPA estimate of 7.4 million USD, rescaled from U.S. to global GDP per capita, for a 2020 value of 2.15 million USD, to reflect the average global GDP per capita level⁵¹. Over time, the VSL grows with the global average GDP per-capita growth rate.

The coefficients $\lambda_{1,c}^{fisheries}$, which represents the market value losses from fisheries, are obtained from a bioeconomic model by Free et al.⁵², which quantified catch and profit levels at the country level for 60% of global fisheries under different management practices and climate impact scenarios. To provide a conservative estimate, we assumed a full adaptation scenario in which maximum sustainable yield is maintained, allowing us to explore how profits under alternative climate scenarios diverge from those under the less climate-impacted RCP 2.6 scenario. The revenue loss at the regional level is then adjusted by a multiplier obtained from Dyck and Sumalia⁵³, which reflects the broader economic impact of fisheries losses on related activities and income.

Finally, the coefficients $\lambda_{1,c}^{ports}$ that accounts for market impacts from climate damages to ports were obtained from an analysis performed by Verschur et al.²⁹, which estimated potential damages to ports due to cyclonic winds, different types of flooding (i.e., coastal, fluvial, and pluvial) and marine extremes (i.e., extreme waves, overtopping, heat and non-cyclonic wind) under climate change scenarios. The damages include direct physical damage to the port infrastructure themselves, representing a financial loss owing to the costs of rebuilding or repairing these infrastructures; operational downtime caused by such damages leads to a decrease in revenue for port operators; and impact on maritime trade activities for countries reliant on these ports. The damage estimates were obtained by fitting a linear regression to the change in damages to ports (as a percentage of GDP) as a function of global mean surface temperature.

For details on all these components, see the SI.

Model simulations to estimate the blue SCC

Based on the Blue RICE50+ model output, we compute the Social Cost of Carbon from oceans and separate it by the different categories and types of ocean services $V_{c,t}^{b,v}$, using the nested utility function (equation 1) and a country-level intertemporal welfare (W) function for each country as discounted utility:

$$W_c = \sum_{t=2020}^{2300} pop_{c,t} \cdot (1 + \delta)^{-(t-2020)} \cdot Utility_{c,t} \quad \text{Eq. 6}$$

Where η is the relative risk aversion. Note that this expression shows undiscounted utility; discounting (eq. 8) is applied in the subsequent aggregation step. We ran the model once (all variables of this run are denoted without hat) and then a second time adding one additional pulse of 1 megaton of CO₂ of emissions in 2020 (variables denoted by a hat). The pulse was added upon a future emissions pathway result of running the model under socioeconomic variables consistent with SSP2 and a no-policy scenario where temperature reaches 3.5 Celsius by the end of the century, equivalent to 6.8 W/m² forcing). We first computed the monetary-equivalent per-capita utility impact at any time t and in country c by computing for each combination of b and v , that is, the difference in utility due to a change only due to impacts on $V_{c,tb,v}$, whereas all other variables are kept at levels of the model run without emission pulse:

$$\Delta Utility_{c,t}^{b,v} = Utility_{b,v,c,t} - \hat{Utility}_{b,v,c,t} \quad \text{Eq. 7}$$

Where $\hat{Utility}$ is the utility result without the pulse and $Utility$ is the utility result with the pulse. This utility change was then converted into monetary equivalents at time t by the marginal utility of one unit of consumption $\frac{\partial Utility_{c,t}}{\partial c_{c,t}}$. Finally, we used the endogenous global Ramsey discount factor, as in Rennert et al. (2022), to convert these values into today's monetary values. The discount factor is computed as:

$$DF_t = (1 + \delta)^{-(t-2020)} \frac{\left(\frac{\sum_c pop_{c,t} c_{c,t}}{\sum_c} \right)^{-\eta}}{\left(\frac{\sum_c pop_{c,2020} c_{c,2020}}{\sum_c} \right)^{-\eta}} \quad \text{Eq. 8}$$

Where the pure rate of time preference is set at $\delta = 0.015$ and the elasticity of marginal utility of consumption at 1.45. Finally, we aggregated all impacts over the total population to get the global value. Thus, we computed the Social Cost of Carbon for 2020 in US-\$[2020] (note that the RICE50+ base currency are US-\$[2005] hence we convert the output in 2020 dollars using the IMF's US-\$ deflator between 2005 and 2020, multiplying all values by 1.298763) based on the sum of the disaggregated impacts due to one ton of CO₂ as:

$$SCC_{2020}^{OCEAN} = \sum_{t=2020}^{2300} (1 + \delta)^{-(t-2020)} \frac{\left(\sum_c \frac{pop_{c,t}}{\sum_{c'} pop_{c',t}} c_{c,t} \right)^{-\eta}}{\left(\sum_c \frac{pop_{c,2020}}{\sum_{c'} pop_{c',2020}} c_{c,2020} \right)^{-\eta}} \left[\sum_c \frac{\Delta Utility_{c,t}}{\frac{\partial Utility_{c,t}}{\partial c_{c,t}}} \cdot \frac{pop_{c,t}}{\sum_{c'} pop_{c',t}} \right] \cdot \sum_c pop_{c,t} \quad \text{Eq. 9}$$

We report SCC values for our central runs as well as the Monte Carlo simulations exploring uncertainties related to the new set of parameters introduced into the model. As an additional test, we also ran a grid of carbon prices verifying that the welfare levels are maximized at the level of the SCC obtained. In the SI, we also explore additional socioeconomic scenarios; moreover, we run a Monte Carlo analysis computing the blue SCC in 10,000 simulations (see SI).

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Code availability

Code for the blue SCC replication can be found at:

github.com/witch-team/RICE50xmodel/releases/tag/v2.6.0 .

The underlying analysis can be found at: github.com/berbastien/blue-scc.

Author Contributions

Except for the first and last authors, all other authors are listed in alphabetical order. BABO: Conceptualization, Investigation, Methodology, Visualization, Formal Analysis, Writing – Original Draft; FG: Methodology, Visualization, Formal Analysis, Software; JE: Methodology, Formal Analysis, Visualization, Software; MT: Methodology, Supervision, Resources; KR: Methodology, Supervision, Writing – Original Draft, Resources, Funding acquisition. All authors contributed equally to Data Curation, Validation and Writing – Review & Editing.

Accounting for Ocean Impacts Nearly Doubles the Social Cost of Carbon

Supplementary Information

This Supplementary Information provides the detailed scientific and economic foundation underlying the parametrization of the blue Social Cost of Carbon (blue SCC). It outlines the data sources, modeling approaches, assumptions, and construction of damage functions used to quantify climate-related welfare impacts across four key ocean-based sectors: corals, mangroves, seaports, and fisheries and mariculture. The final section addresses uncertainty and sensitivity analysis, exploring how blue SCC estimates vary across different climate scenarios and key model parameters.

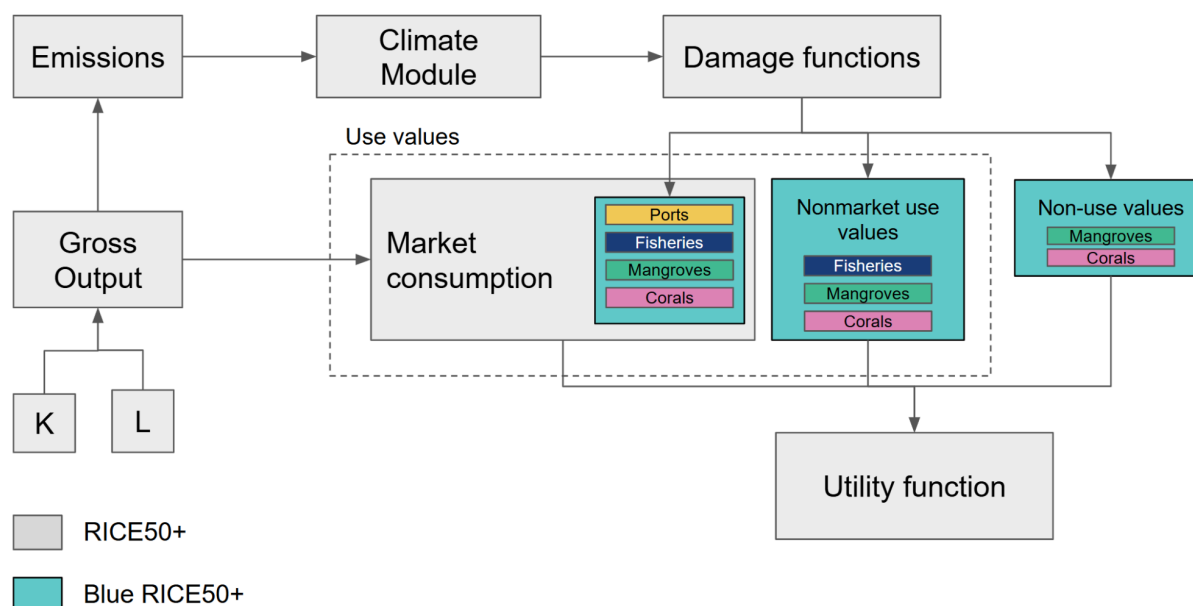


Figure S1. Main components of the Integrated assessment Model RICE50+. This diagram illustrates the key components of the RICE50+ integrated assessment model. Components shown in gray represent the standard structure of RICE50+, including carbon emissions, temperature projections, economic output, and consumption-based welfare calculations. The components shown in blue represent new additions introduced in Blue RICE50+, which incorporate ocean-based damages and their effects on blue well-being.

Corals

Overview

To estimate the economic damages to coral reefs from climate change, we combine two analytical branches that trace the ecological and economic pathways of coral value. The first branch starts with climate impacts, deriving site-level relationships between sea surface temperature and coral cover from Sully et al., which we scale to global mean surface temperature and aggregate spatially to the country level. The second branch focuses on economic valuation, drawing from hundreds of studies in the Ecosystem Services Valuation

Database (Brander et al.), categorizing values into market use, non-market use, and non-use types, and adjusting them for income effects using global income elasticities.

These two branches converge by matching country-level shadow prices with projected coral area and its climate-driven changes. The resulting country-level damage estimates are then integrated into RICE50+, our regionalized integrated assessment model, to quantify the welfare impacts of coral reef degradation under climate change. The overall structure of this workflow is shown in Figure S2.

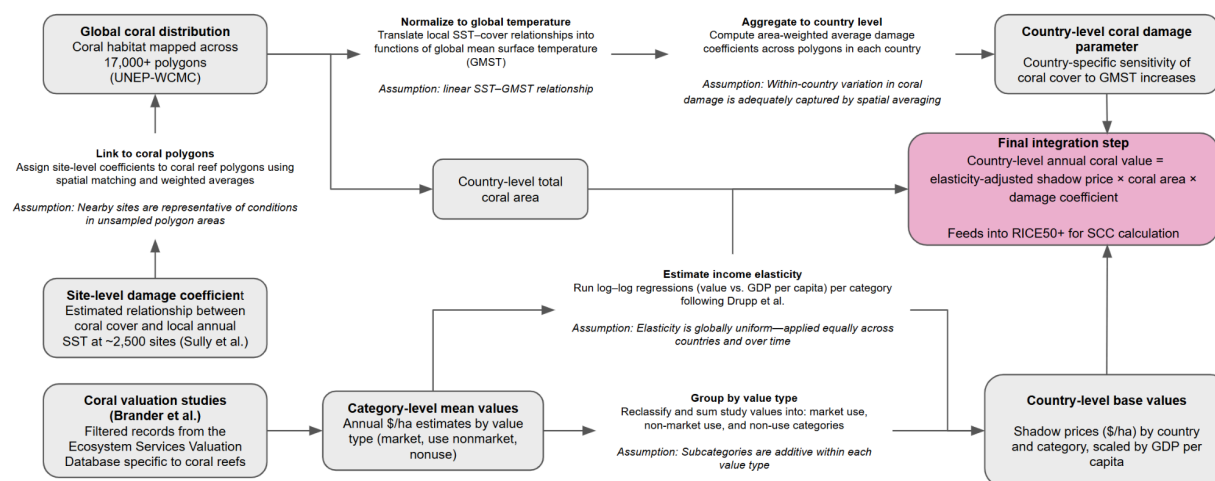


Figure S2. This diagram illustrates the two converging analytical branches used to estimate country-level coral reef damages integrated into the RICE50+ model. The climate impact pathway (top) begins with site-level estimates of coral cover loss in response to sea surface temperature (SST) from Sully et al., which are scaled to global mean surface temperature (GMST), spatially mapped onto global coral reef polygons, and aggregated to country-level damage coefficients. The valuation pathway (bottom) draws from Brander et al.'s coral valuation studies, categorizing values into market use, non-market use, and non-use types, and applying global income elasticity estimates to adjust shadow prices across countries and over time. These two branches converge by combining country-specific coral area, climate sensitivity, and shadow prices to calculate annual losses in ecosystem service value, which are then used as inputs in the RICE50+ integrated assessment model.

Use and Non-use values from Corals

Numerous studies have evaluated both the use and non-use values of coral reefs. Our research requires a comprehensive global assessment, a challenging task given that existing valuations are predominantly localized. One approach adopted by DeGroot et al.⁵⁴ and recently updated by Brander et al. (2024)⁵⁰ addresses this situation by compiling and synthesizing individual studies. Brander et al. used the Ecosystem Services Valuation Database (ESVD)—a substantial collection of original valuation studies across different ecosystems—to assess the magnitude

and variation in economic values of ecosystem services per hectare of 15 biomes, including coral reefs, drawing on insights from 461 individual studies that looked at coral benefits. Table S1 shows the average value of the annual flow of ecosystem services provided by corals, as shown in Brander et al's study.

Table S1. Total coral values per category.

Category	Type of Value	Coral reefs (Int\$/ha/year) [number of studies] (standard error)	Total value (Int\$/ha/year) [number of studies] (standard error)
Food	Use Market	741 [68] (1,321)	19,289 [74] (21783)
Raw materials	Use Market	18,514 [4] (21,743)	
Ornamental resources	Use Market	34 [2] (47)	
Climate regulation	Use Nonmarket	2 [4] (3)	36186 [260] (51218)
Moderation of extreme events	Use Nonmarket	14,369 [17] (44,586)	
Waste treatment	Use Nonmarket	4,078 [8] (2,060)	
Erosion prevention	Use Nonmarket	3,418 [11] (6,819)	
Maintenance of soil fertility	Use Nonmarket	1,551 [3] (9 6 9)	
Aesthetic information	Use Nonmarket	5,580 [11] (1,270)	
Opportunities for recreation and tourism	Use Nonmarket	6,271 [2 0 5] (24,128)	
Inspiration for culture, art and design	Use nonmarket	917 [1] (NA)	

Maintenance of life cycles	Nonuse	1,385 [2] (18)	29610 [115] (56536)
Maintenance of genetic diversity	Nonuse	9,432 [5] (17789)	
Existence, bequest values	Nonuse	18,793 [108] (53,665)	

Our integrated assessment model (IAM), particularly the utility function, is designed to encompass a range of values, including market use values (therefore included in GDP), non-market use values, and non-market non-use values. Consequently, we have reclassified DeGroot's values into three primary categories: market use values, non-market use values and non-use values.

We classified provisioning services such as food, raw materials and ornamental resources as market use values. Non-market use values refer to those services that provide a direct benefit to people although not exchanged in the market, such as climate regulation, moderation of extreme events, waste treatment, and more.

Non-use values encompass the ecological role of corals as habitats for other species and their existence value. Importantly, while some non-use values studies have explicitly estimated values for visitors and non-visitors (19.6%), the large majority of studies have estimated non-use values for the “general public”, “general population”, “residents” and “households” within the country, so we treat non-use values at the national scale. In theory, however, non-use values are not limited by political boundaries—individuals anywhere in the world may derive welfare from knowing that ecosystems like corals exist, regardless of location. Ideally, these values should be reflected globally, possibly weighted by factors such as income or cultural connection. Yet, implementing such a global valuation raises significant challenges, including uncertainty in the spatial distribution of willingness to pay, sociocultural variation, and limited empirical data on how non-use values decay with distance or differ across countries. Moreover, introducing global non-use beneficiaries could dramatically inflate the blue SCC, adding further uncertainty.

For these reasons, and consistent with the available valuation literature, we adopt a conservative approach by restricting non-use beneficiaries to domestic populations only. This aligns with the structure of most underlying studies and avoids introducing unquantified global parameters into the model. We acknowledge that this is a limitation of our current framework and an area for future work—particularly for ecosystems in international waters, such as the deep sea, where defining domestic beneficiaries is inherently problematic.

We write the annual per-area value of these two types of benefits using the notation $p_{c,t}^{coral,v}$, where $v = \{\text{non-market use, non-use}\}$, and t is the year and c the country. In our analysis, we

adjusted the annual per-area values of corals $p_{c,t}^{coral,v}$ based on the per capita gross domestic product (GDP) of countries with coral reefs to allow for the value to increase as society becomes richer. We used the original ESDV studies presented in Brander et al. to estimate the income elasticity of ecosystem service values (γ^v). The values of γ^v for the different cases of v (non-market use, and nonuse) are presented in Table S2. We adjust the ecosystem service value for each country based on a factor that reflects each country's GDP per capita relative to the global mean. This adjustment factor was calculated by raising the ratio of a country's GDP per capita to the mean GDP per capita to the power of γ^v , effectively scaling the base annual per-area value up or down based on the country's relative income. The result is a dataset that not only quantifies the values of coral reefs but also contextualizes these values within the economic landscape of each country, providing a nuanced view of ecosystem valuation across different economic settings.

Table S2. Income elasticities of coral values

Dependent variable: Economic value of corals INT\$_{2020_ha_yr_ln}					
	Coefficient	Std. Error	t	Sig.	
(Constant)	4.615	0.534	8.644	<.001	
GDPPCln_UseMarket	0.129	0.062	2.073	0.039	
GDPPCln_UseNonMarket	0.222	0.058	3.843	<.001	
GDPPCln_NonUse	0.243	0.068	3.575	<.001	
	R	R Square	Adjusted R Square	Std. Error of the Estimate	
	.241 ^a	0.058	0.05	2.39024	
	Sum of Squares	df	Mean Square	F	Sig.
Regression	130.724	3	43.575	7.627	<.001 ^b
Residual	2119.619	371	5.713		
Total	2250.342	374			

In addition to this first adjustment, our analysis also considers that the value of environmental amenities will keep changing with income in the future. Therefore, the income elasticity should be applied once again to compute the future per-area values of corals, as follows:

$$p_{c,t+1}^{coral,v} = p_{c,t}^{coral,v} \cdot (1 + \gamma^v \cdot g_{c,t}) \quad \text{Eq. C1}$$

Where $g_{c,t}$ is the GDP per capita growth. The total country-level use and non-use values provided by coral reefs in a year are, therefore function of the per-area values given in equation C1, and the total coral area at any given year ($A_{c,t}^{coral}$), in the following way:

$$V_{c,t}^{coral,v} = p_{c,t}^{coral,v} \cdot A_{c,t}^{coral} \quad \text{Eq. C2}$$

The coral area, in turn, is affected by climate change as described in the following section.

Climate impacts on corals

In a comprehensive study, Sully et al.²⁷ analyzed the impact of climate change on coral reefs. They gathered over 7,000 surveys, conducted from 1997 to 2018, which covered coral conditions at 2,949 different sites. The aim was to understand how 14 local environmental factors influenced coral cover. Employing a hierarchical Bayesian model, their analysis revealed a significant negative impact of mean sea surface temperature on coral cover. The results from this model, coupled with climate model projections consistent with the RCP4.5 and RCP8.5 scenarios, enabled them to project changes in coral cover at each of these sites for two future time periods: 2050 and 2100.

In our research, we aim to estimate the relationship between global mean surface temperature (GMST) and changes in coral cover $A_{(i)}$ in each site (i), so we re-scale Sully et al. results. First, we extract the projected coral cover data for each site from the study by Sully et al. This data consists of four projections for each site, derived from two climate scenarios (RCP4.5 and RCP8.5) and two time horizons (2050 and 2100). We then augment this dataset with information on the corresponding global mean temperature change relative to the same baseline, for each time horizon and scenario ($\Delta GMST_t$). With this expanded database, we calculated the coefficient (δ_i^{coral}) of a linear function representing the change in coral cover as a function of global mean surface temperature (Figure S2, left panel), as shown in the equation below:

$$A_{i,t} = \alpha + \delta_i^{coral} \Delta GMST_t + \epsilon_t \quad \forall i \quad \text{Eq. C3}$$

This approach, while straightforward, was crucial for integrating these findings into our broader research. Our work primarily involves an Integrated Assessment Model (IAM) that operates on the basis of global or regional mean temperatures, but we cannot account for local sea surface temperatures near each coral reef site, which are critical in the original study. Therefore, by correlating global mean temperature with coral cover change, we can effectively utilize these insights within the context of our IAM-focused research.

The coefficients (δ_i^{coral}) obtained are shown in the right panel of Figure S3, plotted against the fraction of the coral reef that is covered by living corals at each site in present day. We observe a slight positive relationship indicating that sites where living corals cover a smaller fraction of

the reef will be more affected by an increase in global mean surface temperature. This is likely to play a role in the distribution of damages.

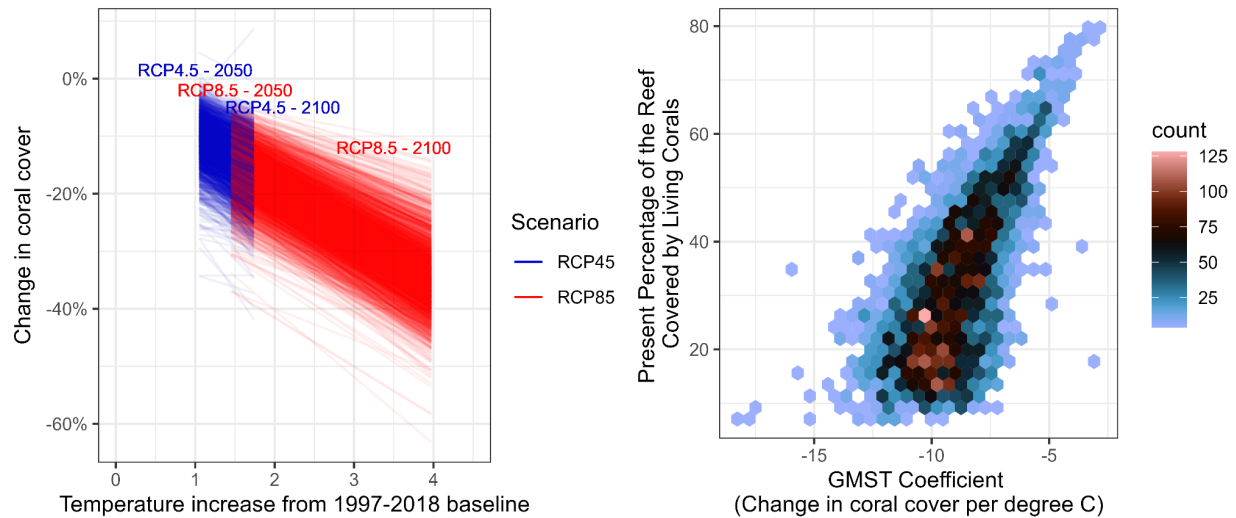


Figure S3. Coral cover damage as a function of global mean surface temperature. Left panel: Linear relationship between projected coral cover loss and global mean surface temperature (GMST), derived from Sully et al. (2019). Right panel: Site-level coral damage coefficients plotted against present-day coral cover fraction at each location. A slight positive trend indicates that sites with lower current coral cover are projected to experience greater sensitivity to warming.

The next step was to bridge the information on the climate impacts on coral reefs at specific survey sites (spatial points) with the actual coral-covered areas worldwide (polygonal areas). This is particularly important because most of the ecosystem services valuation literature, as seen in the previous subsection, is given in terms of value per area. To solve this challenge, we used the geographic dataset from the UNEP World Conservation Monitoring Centre, which comprises over 17,000 spatial polygons representing coral-covered areas across the globe. This comprehensive dataset allowed us to establish a link between the individual site-based climate impact findings and the global distribution of coral reefs.

To integrate our findings with this global perspective, we aligned both the survey datapoints and the UNEP coral reef area distribution dataset within the same coordinate system. This alignment facilitated a direct comparison and overlap of data, enhancing our understanding of how climate impacts at point locations translate to potential impacts to coral-covered areas. This process and the resulting integrated data representation can be seen in Figure S4, providing a visual reference for this crucial step in our analysis.

Coral Area and Cover Change

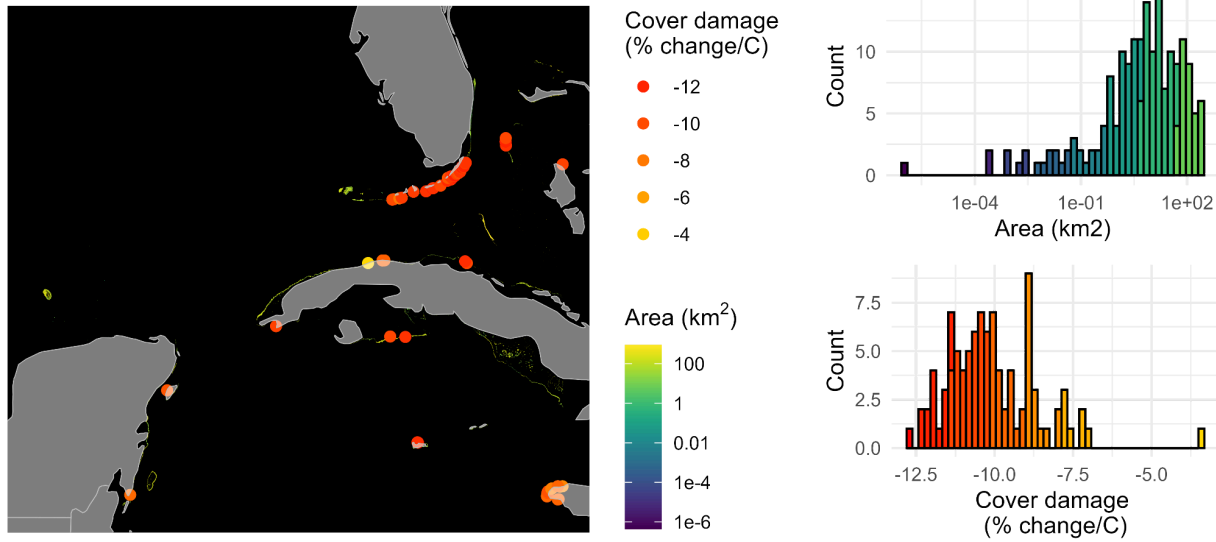


Figure S4. Alignment of coral survey sites with global coral reef polygons. Left panel: spatial integration of coral condition survey points from Sully et al. with the global coral reef polygon dataset from UNEP-WCMC. Right top panel: frequency of the area coverage across the polygons shown in the left. Right bottom panel: frequency of damage coefficient magnitude across the points shown in the left.

Not all polygons in the WCMC dataset had a corresponding data point from which to derive the climate impact coefficient. To solve this, we adopted the following methodology. For polygons without any associated survey data points, we assigned the average of the three nearest coefficients based on Euclidean distance. For polygons with a single survey data point within their area, we directly assigned the coefficient from that specific survey point. And in cases where a polygon encompassed more than one survey data point, we calculated the average value of all the data points within that area. This averaging accounted for the variability within larger areas, offering a more comprehensive view of the climate impact across different parts of the same coral-covered area. This methodology allowed us to effectively map the climate impact coefficients onto the global coral reef distribution, thus linking localized climate impact data with the broader context of global coral reef distribution and their associated values.

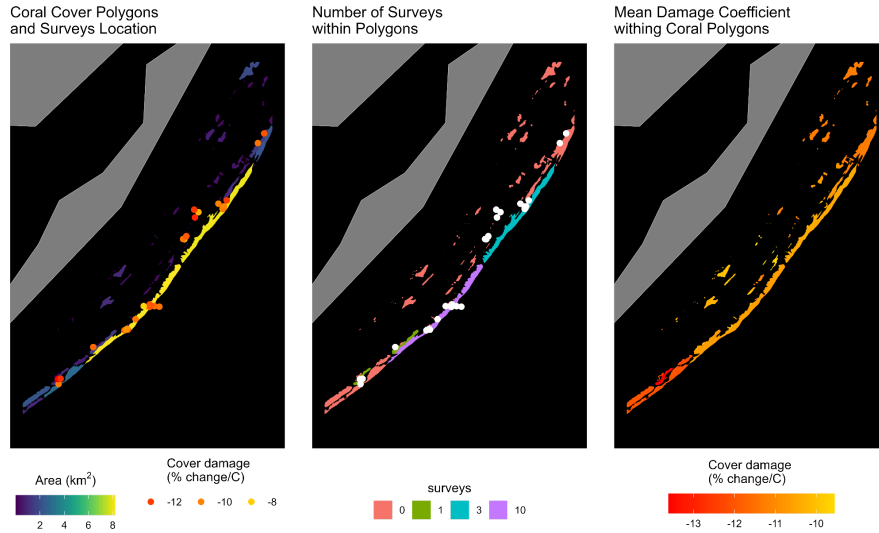


Figure S5. Application of climate impact assignment methodology to coral reefs near the Florida Keys. This figure provides a regional example of how climate impact coefficients are assigned to coral reef polygons using spatial interpolation from site-level survey data. Left panel: Coral polygons and damage points showing their respective information. Middle panel: number of coral damage surveys (points) within each polygon. Right: assignment of damage coefficients to coral polygons.

We included Figure S5 which shows coral reefs near the Florida Keys, to illustrate the application of our methodology for assigning climate impact coefficients to coral reefs. Figure S6 expands on this by presenting the global distribution of coral-covered areas, each colored with their assigned cover damage coefficient. This coefficient represents the estimated fraction of coral cover loss corresponding to an additional degree of warming of the global mean temperature.

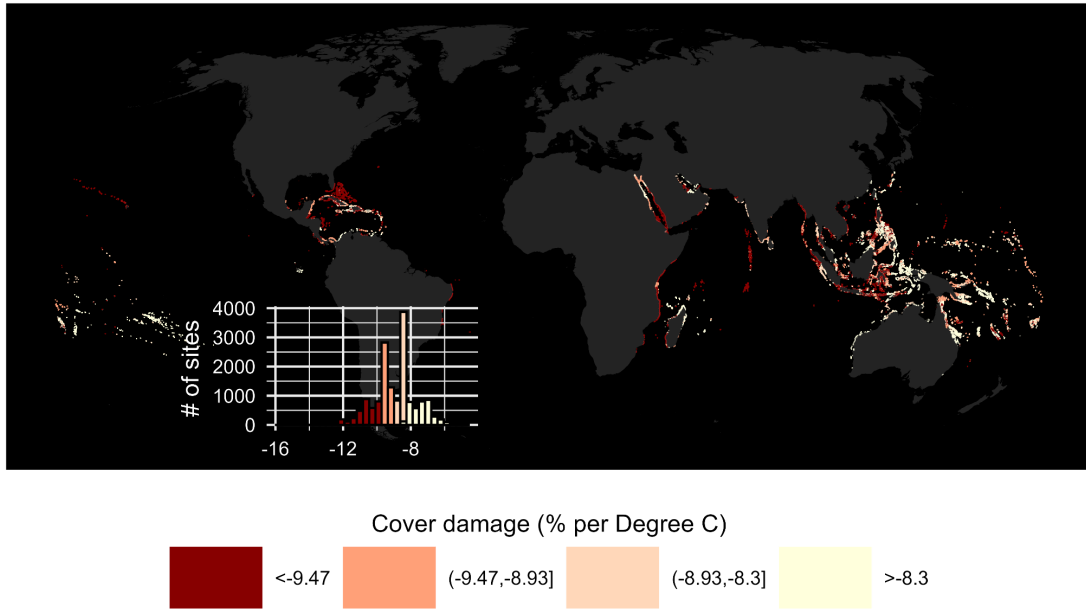


Figure S6. Global distribution of coral damage coefficients. Global map of coral-covered areas, with each polygon colored according to its assigned climate impact coefficient, divided into four quantile groups.

Finally, we aggregated the data to the country level. This step was essential for integrating our analysis within the framework of RICE50+, which uses countries as its smallest jurisdictional unit. We calculated the coral-area-weighted average damage coefficient for each country (δ_c^{coral}). This involved averaging the damage coefficients assigned to all coral spatial polygons $i(c)$ within a country's c jurisdiction, but with the crucial distinction that the area of each coral polygon was used as a weight for its coefficient. This means that coefficients assigned to larger coral areas carried more weight in this country-level aggregation:

$$\delta_c^{coral} = \sum_{i(c)} \frac{A_{i(c)} \cdot \delta_i^{coral}}{\sum_{i(c)} A_{i(c)}} \quad \text{Eq. C4}$$

The results are presented in Figure S7.

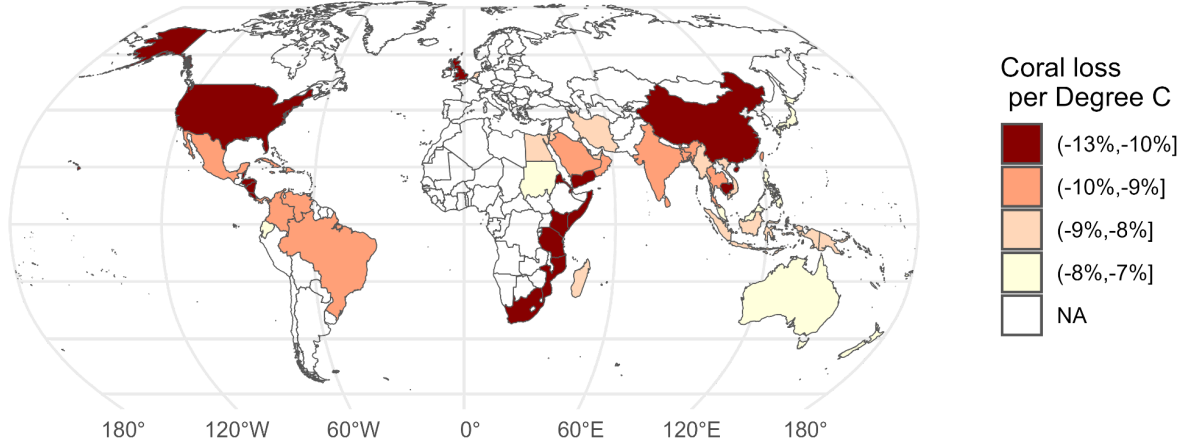


Figure S7. Country-level coral damage coefficients. Area-weighted average damage coefficients for each country shown as percentage of coral area loss per degree of warming, based on the climate impacts to coral-covered areas within national jurisdictions.

We use the coefficients θ_c (Extended Data Table 1) to project changes in coral cover area ($A_{c,t}^{coral}$) as a function of global mean surface temperature (T):

$$A_{c,t}^{coral} = A_{c,0}^{coral} \cdot (1 + \delta_c^{coral} \Delta T_t) \quad \text{Eq. C7}$$

Corals market damages

To estimate the market damages $D_{c,t}^{coral}$ of climate impacts on coral reefs, we first estimate the market benefits from corals at the country level under damaged and non-damaged scenarios, using, on the one hand, the per-area marginal value of corals in Table S1 adjusted by the income elasticity as in equation C1 and, on the other hand, the damage estimate coefficients that capture the temperature effects on coral area cover (Eq. C7) as follows:

$$D_{c,t}^{coral} = \frac{p_{c,t}^{coral} (A_{c,t}^{coral} - A_{c,0}^{coral})}{GDP_{c,t}} \quad \text{Eq. C9}$$

$$= \frac{p_{c,t}^{coral} (A_{c,0}^{coral} \cdot (1 + \delta_c^{coral} \Delta T_t) - A_{c,0}^{coral})}{GDP_{c,t}} \quad \text{Eq. C10}$$

$$= \frac{p_{c,t}^{coral} (A_{c,0}^{coral} \cdot ((1 + \delta_c^{coral} \Delta T_t) - 1))}{GDP_{c,t}} \quad \text{Eq. C11}$$

$$= \frac{p_{c,t}^{coral} (A_{c,0}^{coral} \delta_c^{coral} \Delta T_t)}{GDP_{c,t}} \quad \text{Eq. C12}$$

$$= \mu_{c,t}^{coral} \cdot \Delta T_t \quad \text{Eq. C12}$$

Where $\mu_{c,t}^{coral} = \frac{p_{c,t}^{coral} A_{c,0}^{coral} \delta_c^{coral}}{GDP_{c,t}}$, We call λ_c^{coral} the market damage coefficient as given by

$\lambda_c^{coral} = \sum_{t=1}^{\tau} \frac{\mu_{c,t}^{coral}}{\tau}$. The standard error of λ_c^{coral} , called $\sigma_{\lambda_c^{coral}}$ is given by the expression

$$\sigma_{\lambda_c^{coral}} = \sqrt{\left(\frac{p_{c,t}^{coral} A_{c,0}^{coral} \Delta T_t}{GDP_{c,t}} \right)^2 \cdot \sigma_{\delta_c^{coral}}^2}, \text{ following the standard rules for uncertainty calculation}^{55}. \text{ Figure}$$

S8 (bottom) shows the fitted lines using λ_c^{coral} compared to the estimated damages as given by equation C9.

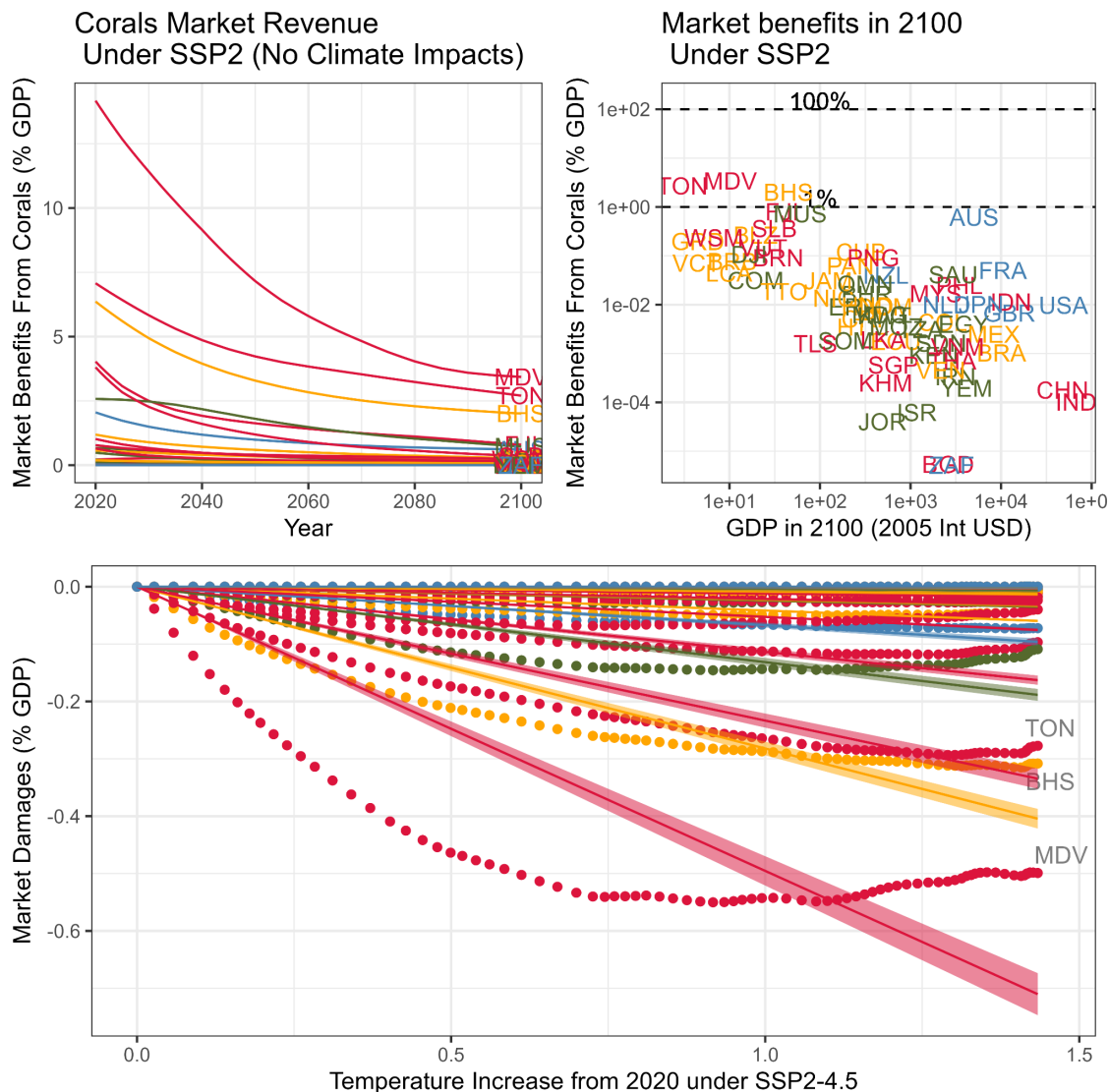


Figure S8. Coral market benefits and climate-related losses. Top left: Projected market benefits from coral reefs under a scenario without climate impacts. Top right: Projected coral market benefits in the year 2100 including climate impacts. Bottom: Estimated differences in coral market benefits with and without climate impacts (dots), along with the fitted linear damage function (lines) and its 95% confidence interval.

We also evaluated a quadratic mapping from GMST to coral market damages, but the estimated fit became concave over part of the model's temperature support, implying decreasing marginal damages at higher GMST that led to solver instability in optimization runs. We therefore retain a monotone linear GMST specification for transparency and stability. This functional-form choice applies to the coral market-use channel; this channel contributes <1% to the central Blue SCC, so the aggregate results are insensitive to this specification.

Mangroves

Overview

Mangroves provide multiple benefits to people.^{56,57} To estimate the welfare losses from mangrove degradation under climate change, we build on the high-resolution framework developed by Bastien-Olvera et al.²⁸ Their approach mirrors the coral analysis in structure but applies a finer spatial scale: mangrove values and damage functions are first estimated at the grid-cell level, using local socioeconomic predictors (notably, nearby GDP per capita) and grid-level climate responses. Local value estimates are based on a meta-regression informed by Brander et al.'s income elasticity findings, allowing projections of per-hectare shadow prices over time. Climate impacts on mangrove area are similarly derived at the grid scale using local sea surface temperature of the hottest month and then aggregated to the country level, with damage functions normalized to global mean surface temperature to maintain consistency with the broader modeling framework. These two pathways—value and area dynamics—converge at the country scale, where they are integrated into the Blue RICE50+ model to quantify the welfare consequences of mangrove loss. The full workflow is illustrated in Figure S9.

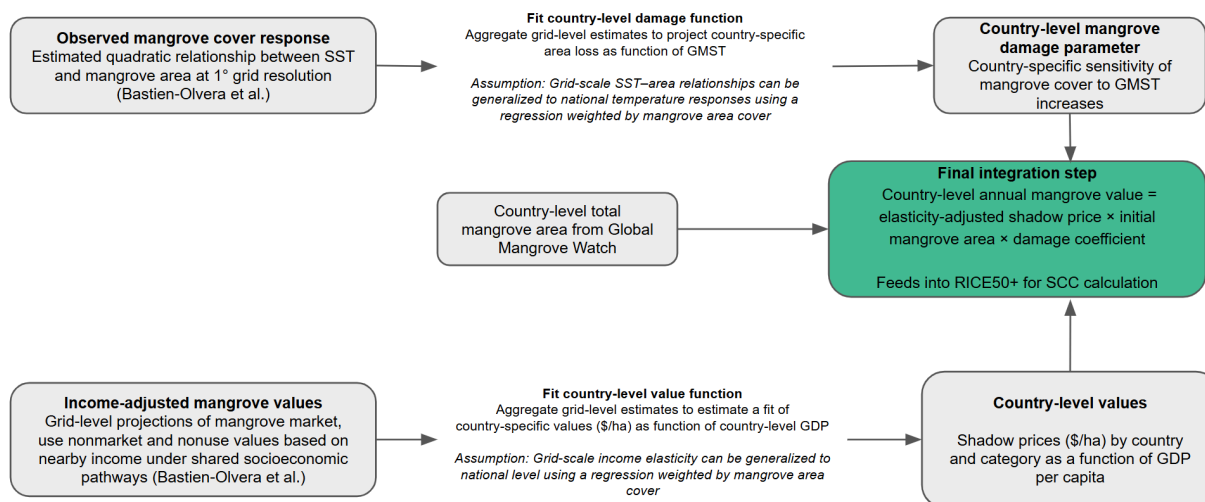


Figure S9. Workflow for estimating climate damages to mangrove ecosystems.

This diagram summarizes the two converging analytical pathways used to quantify mangrove-related welfare losses under climate change. The climate impact pathway (top) draws on grid-level damage estimates from Bastien-Olvera et al., which relate mangrove area loss to local sea surface temperatures and are normalized to global mean surface temperature before aggregating to the country level. The valuation pathway (bottom) estimates per-hectare mangrove values using a regression informed by local GDP per capita and income elasticity parameters from Brander et al. These values are projected over time and combined with projected mangrove area losses to estimate annual economic damages at the country level. The resulting impacts are incorporated into the Blue RICE50+ model to assess changes in global welfare under future climate scenarios.

Climate Impacts on Mangrove Area

Mangrove forests will be negatively impacted by increasing temperatures worldwide. While some colder places might see benefits in mangroves due to additional warming, it is expected that mangroves, as any other ecosystem, have a temperature threshold above which their coverage will decline. Bastien-Olvera et al.²⁸ used global high-resolution data of mangrove cover from 1996 to 2020 to estimate the effects of temperature increase in mangrove cover. Their preferred specification uses the sea surface temperature of the hottest month in its quadratic form as the main climate-forcing variable that impacts mangrove coverage at the local level using 1-by-1-degree grid cells. Here, we use such results to estimate a country-level damage function that projects changes in mangrove cover as a function of global mean surface temperature (T):

$$A_{c,t}^{mangroves} = A_{c,t}^{mangroves} \left(1 + \delta_{1,c}^{mangroves} \cdot \Delta T_t + \delta_{2,c}^{mangroves} \cdot \Delta T_t^2 \right) \quad \text{Eq. M1}$$

The estimated mangrove area in 2020 and the damage function coefficients are provided in Extended Data Table 2 and shown in Figure S10:

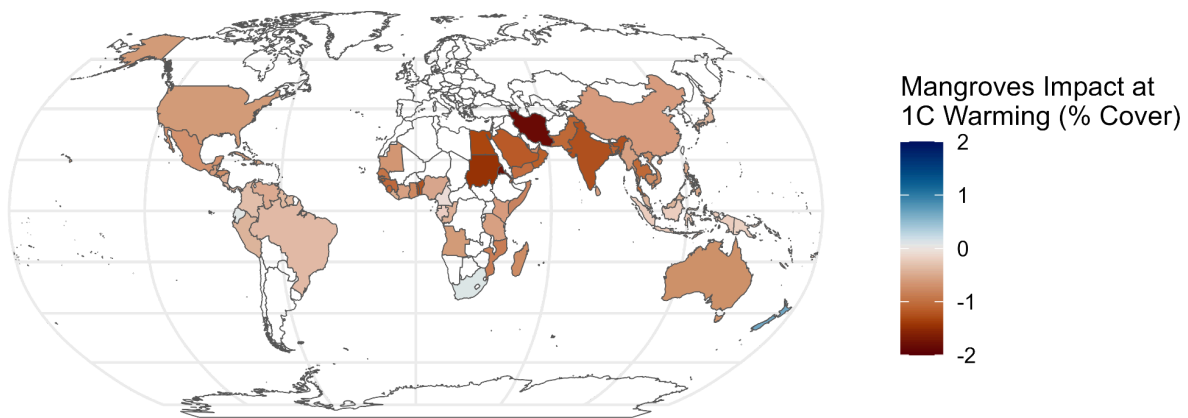


Figure S10. Mangrove cover damage under 1°C of global warming. Country-level projections of mangrove cover change in response to a 1°C increase in global mean surface temperature, based on grid-level estimates from Bastien-Olvera et al.

Mangroves Non-Market Use and Non-Use Values

Mangroves close to human settlements or economically active areas tend to offer more benefits to people than those located further away. Bastien-Olvera et al. estimated the value of the benefits provided by a hectare of mangrove for more than 1500 locations across the world, considering this heterogeneity in the per-area values. Spatially explicit global trajectories of

GDP and population consistent with the shared socioeconomic pathways were used as predictor variables of the per-area value of mangroves. The specific coefficients that allowed to project the per-area values as a function of local population and GDP were calculated using a meta-regression model approach that considered 371 (118; 96) primary studies on provisioning (regulating; cultural) services of mangroves registered in the TEEB Ecosystem Services Valuation Database⁵⁴. Descriptive statistics of the original studies in the ESVD that we used are shown in Table S3.

Supplementary Table S3. Total mangrove values per category.

Ecosystem Service	Type of Value	ES value (Int\$/ha/year)	Total value (Int\$/ha/year)
		[number of studies]	[number of studies]
		(standard error)	(standard error)
Food	Use Market	3,595	5,825
		[252]	[371]
		(707)	(1,547)
Raw materials	Use Market	2,230	
		[119]	
		(840)	
Climate regulation	Use Nonmarket	2,997	17,600
		[43]	[118]
		(1702)	(2,122)
Moderation of extreme events	Use Nonmarket	2448	
		[32]	
		(704)	
Waste treatment	Use Nonmarket	4,557	
		[18]	
		(1,785)	
Erosion prevention	Use Nonmarket	6,570	
		[20]	
		(2,919)	
Maintenance of soil fertility	Use Nonmarket	1,028	
		[5]	
		(438)	
Aesthetic information	Use Nonmarket	30,340	10198
		[4]	[96]
		(30,006)	(12,121)
Opportunities for recreation and tourism	Use Nonmarket	3,351	
		[67]	
		(1,068)	

Inspiration for culture, art and design	Use nonmarket	3,890	
		[1]	
		(NA)	
Information for cognitive development	Use nonmarket	749	
		[6]	
		(486)	
Existence, bequest values	Nonuse	12,660	
		[18]	
		(7,870)	

Bastien-Olvera et al. provided global, grid-level estimates of mangrove per-area benefits, adjusting for changes in socioeconomic factors near mangroves throughout the century as given by global gridded shared socioeconomic pathways. In our study, we compile these benefits to create a time-series area-weighted average shadow prices for mangroves ($p_{c,t}^{mangroves,v}$) for three types of values (v), for each country (c) across all years (t) throughout the century. We perform a regression to estimate how this average value would change over time as a function of the country-level GDP per capita ($GDPpc$). For each country c and for each value v provided by mangroves (non-market use values and non-use values), we estimate the following regression:

$$\log(p_t^{mangroves,v}) = a^v + b^v \log(GDPpc_{c,t}) + \epsilon_t \quad \forall c, v \quad \text{Eq. M2}$$

The coefficients a and b , for each country and type of value, are provided in Extended Data Table 3. Therefore, ocean-based values ($V_{c,t}^{mangroves,v}$) provided by mangroves in a country c in year t are given by multiplying the mangrove area ($A_{c,t}^{mangroves}$) in the country by the per-hectare value:

$$V_{c,t}^{mangroves,v} = p_t^{mangroves,v} \cdot A_{c,t}^{mangroves} \quad \text{Eq. M3}$$

Mangroves Market Values

To estimate market damages $D_{c,t}^{mangroves}$ stemming from climate impacts on mangroves, we first calculate the value of the area lost as a consequence of climate change:

$$D_{c,t}^{mangroves} = \frac{p_{c,t}^{mangroves} (A_{c,t}^{mangroves} - A_{c,0}^{mangroves})}{GDP_{c,t}} \quad \text{Eq. C9}$$

$$= \frac{p_{c,t}^{coral} (A_{c,0}^{mangroves} \cdot (1 + \delta_{1,c}^{mangroves} \Delta T_t + \delta_{2,c}^{mangroves} \Delta T_t^2) - A_{c,0}^{mangroves})}{GDP_{c,t}} \quad \text{Eq. C10}$$

$$= \lambda_{1,c,t}^{mangroves} \cdot \Delta T_t + \lambda_{2,c,t}^{mangroves} \Delta T_t^2 \quad \text{Eq. C11}$$

Where $\lambda_{1,c,t}^{mangroves} = \frac{p_{c,t}^{mangroves} \cdot A_{c,0}^{mangroves} \cdot \delta_{1,c}^{mangroves}}{GDP_{c,t}}$, and $\lambda_{2,c,t}^{mangroves} = \frac{p_{c,t}^{mangroves} \cdot A_{c,0}^{mangroves} \cdot \delta_{2,c}^{mangroves}}{GDP_{c,t}}$.

The standard error of $\lambda_{n,c}^{mangroves}$ for $n \in \{1, 2\}$, called $\sigma_{\lambda_{n,c}^{mangroves}}$ is given by the expression

$$\sigma_{\lambda_{n,c}^{mangroves}} = \sqrt{\left(\frac{p_{c,t}^{mangroves} \cdot A_{c,0}^{mangroves} \cdot \Delta T_t^n}{GDP_{c,t}} \right)^2 \cdot \sigma_{\delta_c^{mangroves}}^2}, \text{ following the standard rules for uncertainty}$$

calculation⁵⁵. Figure S11 (bottom) shows the fitted lines using $\lambda_{n,c}^{mangroves}$ compared to the estimated damages as given by equation C9.

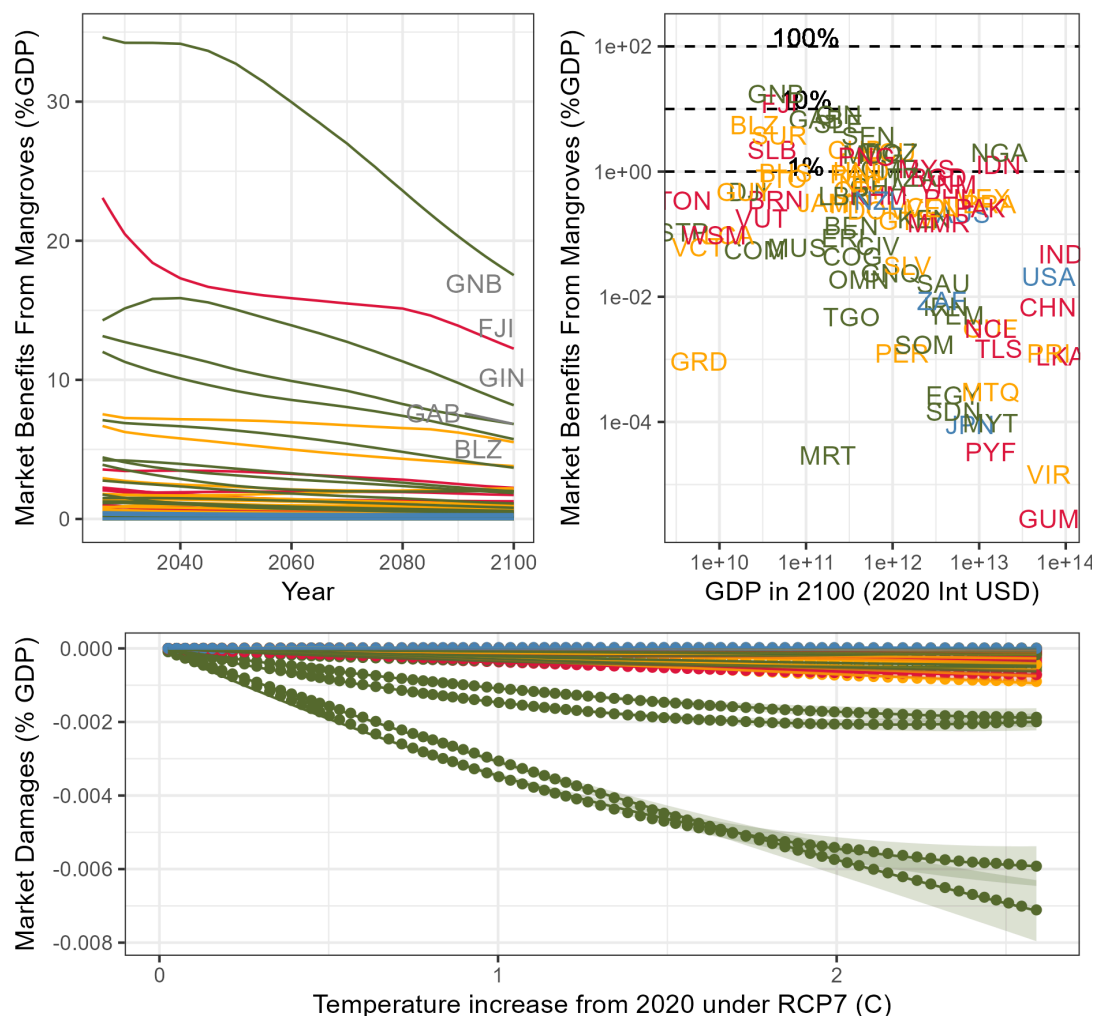


Figure S11. Mangrove market benefits and climate-related losses. Top left: Projected market benefits from mangroves under a scenario without climate impacts. Top right: Projected mangrove market benefits in the year 2100. Bottom: Country-level market damages from mangrove loss under a high warming scenario (RCP7.0). Dots represent model-based damage estimates, while the solid line shows the fitted quadratic damage function with its 95% confidence interval (shaded area).

Ports

Overview

To estimate the damages from climate change on maritime trade infrastructure, we focus on projected economic risks to ports from increased flooding and cyclonic winds^{29,58}. Site-level estimates covering approximately 1,300 ports worldwide combine projections of future climate hazards with expected economic growth under multiple SSP scenarios. We express future port-related economic risks as a share of country-level GDP and develop country-specific

damage functions by regressing changes in risk against projected increases in global mean surface temperature. This approach captures both physical exposure and socioeconomic dynamics while assuming no future adaptation, rerouting, or construction of new ports. However, it allows for local expansion of existing port infrastructure as trade volumes increase. Verschuur and colleagues²⁹ projected growth in trade derived from a computable general equilibrium (CGE) model and allocated it across five major port types—container, dry bulk, liquid bulk, Ro-Ro, and general cargo—based on standard scaling ratios of cargo volume per unit area. Specifically, the following average trade-to-area conversion factors are used: 4.1 tonnes/m² for Ro-Ro, 7.5 tonnes/m² for general cargo, 33.1 tonnes/m² for liquid bulk, 19.4 tonnes/m² for dry bulk, and 17.3 tonnes/m² for container ports⁵⁹. Future climate impacts were estimated under these assumptions⁶⁰. For transparency, Extended Data Table 4 reports the projected trade value handled through ports in 2030 and 2050 under SSP1, SSP2, and SSP5. The resulting country-level damages are integrated into the Blue RICE50+ model to assess how climate impacts on ports affect global welfare. The full workflow is summarized in Figure S12.

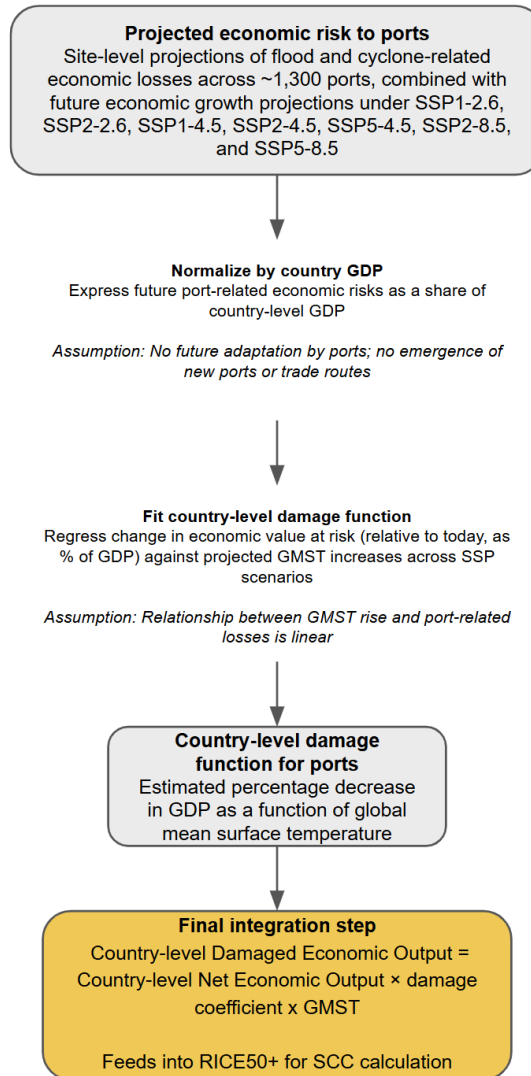


Figure S12. This diagram outlines the steps used to estimate country-level damages to ports from climate change. Site-level projections of flood and cyclone-related risks to ~1,300 ports are combined with economic growth forecasts under seven shared socioeconomic pathways (SSPs). Future economic value at risk is normalized by country-level GDP, assuming no adaptation or emergence of new ports. A country-specific damage function is then derived by regressing increases in economic risk (as a share of GDP) against global mean surface temperature rises. These damage functions are incorporated into the Blue RICE50+ model to quantify welfare impacts of port-related climate risks under future warming scenarios.

Damage Function

Ports are critical infrastructure that handles 50% of global trade in terms of value⁵⁸. Extreme weather events, exacerbated by climate change, have the potential to affect seaports that are crucial for international trade. Verschuur et al.²⁹ quantified the economic risk for 1320 ports worldwide due to cyclonic winds, different types of flooding (coastal, fluvial, and pluvial) and marine extremes (i.e., extreme waves, overtopping, heat and non-cyclonic wind). The

implications of these extreme events at ports are multifaceted. First, there is the direct physical damage to the port infrastructure themselves, representing a financial loss owing to the costs of rebuilding or repairing these infrastructures. Second, operational downtime caused by such damages leads to a decrease in revenue for port operators. Additionally, there is an impact on maritime trade activities for countries reliant on these ports. The impact thus extends beyond the port boundaries and looks at economic spillovers, or systemic impacts, to countries. In order to quantify the risk to trade flows, a coupled trade-transport model is required, which maps out which ports are used for which bilateral trade flows⁵⁸. This thus captures how operations downtime at port can cause impacts to trade flows, both in terms of the value of imported and exported goods.

These three loss-components cover the economic value at-risk (EVaR). Overall, the largest fraction of the economic risks are related to maritime trade activities (90%), followed by physical impacts (8%) and loss of revenue at the ports (2%). Figure S13 A shows the present total estimated economic value at-risk per country as a percentage of its GDP. On average, lower-income economies are more vulnerable to climate-related losses to ports.

Verschuur et al.²⁹ conducted a study estimating the expected change in economic value at risk for 1,320 ports under various cyclone wind and flooding conditions projected for 2050. These projections are based on Representative Concentration Pathways (RCPs) 2.6, 4.5, and 8.5, alongside different scenarios of port-related economic activity growth linked to Shared Socioeconomic Pathways (SSPs) 1, 2, and 5. The analysis assumes no adaptation measures are implemented by ports; thus, future EVaR estimates scale with projected hazard risks (i.e., increases or decreases in physical asset damages and downtime) and trade volume growth, which impacts the flow of trade through ports and the need for infrastructure expansion. Figure S13 B displays the mean projected future EVaR, averaged across the seven RCP-SSP combinations, indicating that while countries' GDP is expected to grow by 2050, the projected EVaR as a percentage of GDP is also anticipated to rise.

Notably, not all countries experience an increase in the EVaR from the present to future conditions. Figure S13 C illustrates the density distribution of changes in the EVaR as a percentage of GDP by 2050 compared to current conditions. Initially, a majority of countries are projected to face an increased EVaR in the future, with only a small portion of the distribution showing a reduction. Furthermore, EVaR is projected to escalate under warmer climate scenarios (across RCP2.6, 4.5, and 8.5) and various SSPs. This increase is attributed to more intense climate change, which is likely to result in more severe winds and flooding. Additionally, strong socioeconomic growth scenarios, characterized by rapid accumulation of material wealth and increased trade, are expected to increase the amount of infrastructure, revenues, and trade exposed to climate extremes.

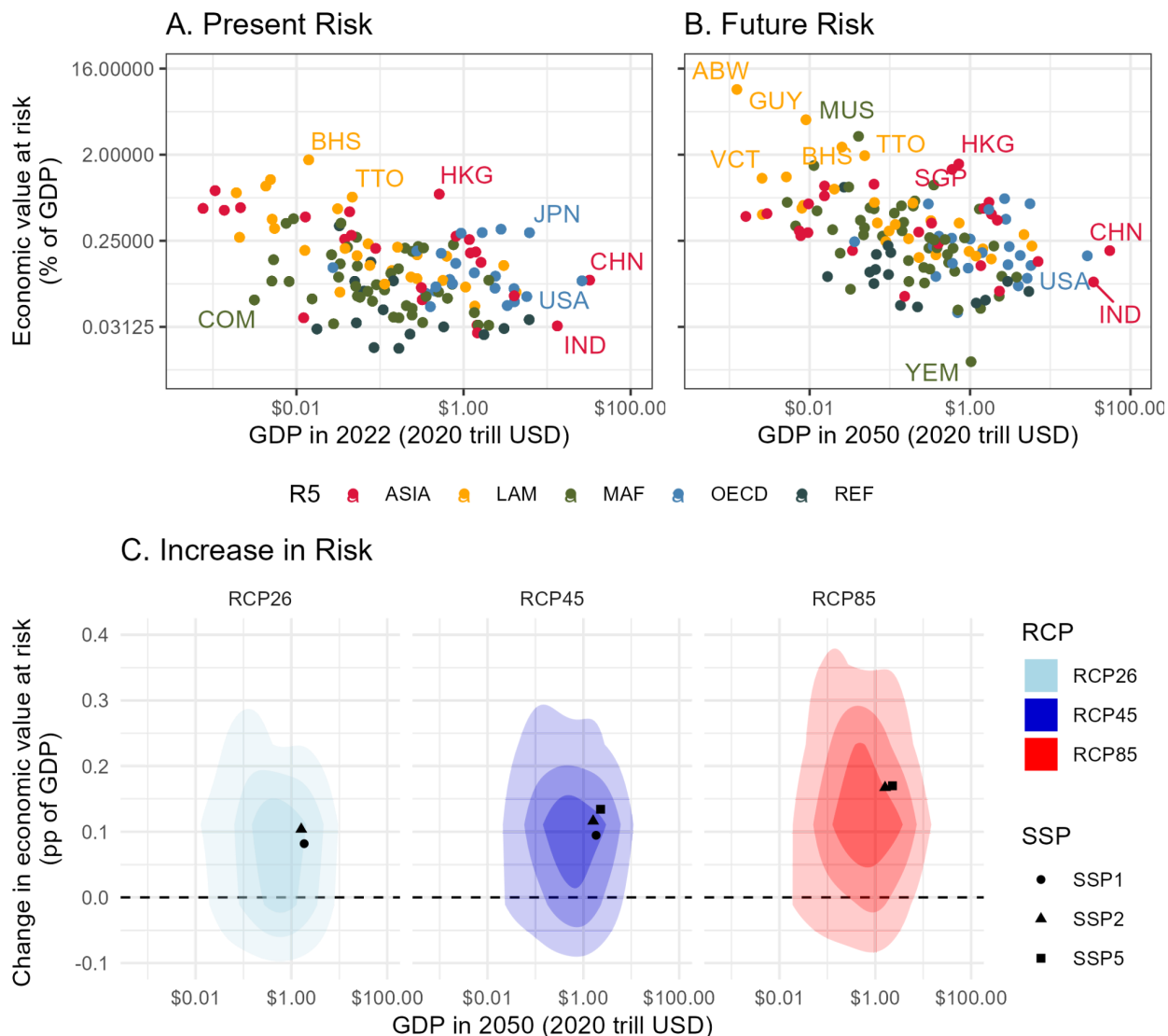


Figure S12. Present and future port-related economic value at risk as a percentage of GDP. Panel A: Current estimated economic value at risk from cyclonic winds, flooding, and marine extremes for each country, expressed as a share of GDP. Panel B: Projected economic value at risk for 2050, averaged across seven SSP–RCP combinations, assuming no future adaptation. Panel C: Distribution of changes in economic value at risk from present to 2050 across countries, shown as differences in percentage of GDP.

The distribution of future EVaR varies significantly across countries, as many of the world's poorest nations are more dependent on port-related economic activities, as indicated by their higher reliance on maritime trade and elevated trade-to-GDP ratios. We compute the change in EVaR from today to 2050 as a fraction of GDP under different shared socioeconomic pathways (Eq. P1):

$$D_{c,2050,ssp}^{ports} = \frac{EVaR_{c,2050,ssp}}{GDP_{c,2050,ssp}} - \frac{EVaR_{c,0}}{GDP_{c,0}} \quad \text{Eq. P1}$$

Figure S14 A depicts the smoothed relationship between the countries' GDP in 2050 and the change in EVaR as percentage of GDP. Although projections for individual countries show variability – including a few small economies that might see a slight increase in EVaR or even a decrease, due to GDP growth outpacing EVaR growth–, the overall trend suggests that smaller economies are expected to face larger increases in EVaR. On average, the poorest nations could see their EVaR as percentage of GDP increase by approximately 2 to 4 percentage points. In contrast, wealthier countries are likely to experience much smaller changes in their EVaR changes compared to their GDP.

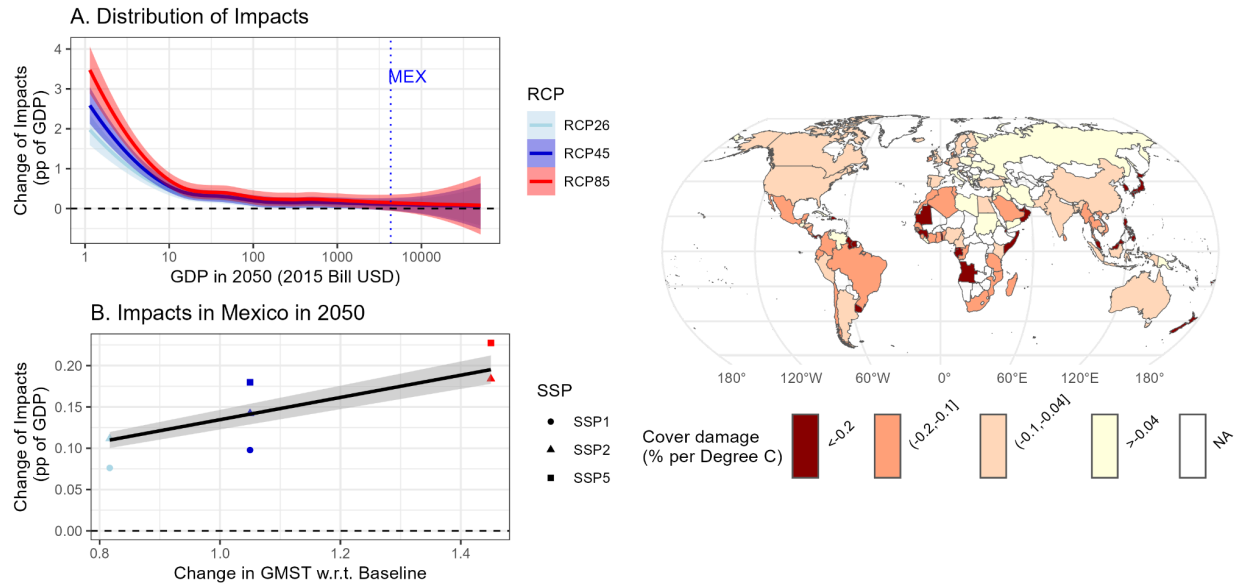


Figure S14. Distribution of port-related damages across countries and warming levels. Panel A: Smoothed relationship between projected 2050 GDP and the change in port-related economic value at risk (EVaR) as a percentage of GDP. The figure highlights that lower-income countries tend to face larger proportional increases in EVaR. Panel B: Example from Mexico showing the relationship between country-level port damages ($\Delta EVaR$ as % of GDP) and global mean surface temperature increase across seven SSP–RCP combinations. A linear regression through the origin is fitted to derive the country-specific damage coefficient used in the model, presented for all countries in the map on the right.

For each country, we summarize the relationship between economic damages to ports ($D_{c,2050,ssp}^{ports}$) and the increase in global mean surface temperature ($\Delta T_{2050,ssp}$), compared to present-day temperatures. We use the point estimates for each country under SSP1-2.6, SSP2-2.6, SSP1-4.5, SSP2-4.5, SSP5-4.5, SSP2-8.5, and SSP5-8.5 to fit a linear regression

model that passes through the origin. An example for this analysis for Mexico is shown in Figure S11B. We apply this method for every country (c) to derive a unique damage function coefficient ($\lambda_{1,c}^{ports}$):

$$D_{c,2050,ssp}^{ports} = \lambda_{1,c}^{ports} \Delta T_{2050,ssp} + \epsilon_{ssp} \quad \forall c \quad \text{Eq. P2}$$

The $\lambda_{1,c}^{ports}$ coefficients (Extended Data Table 5) bring port-related EVaR from climate change into the RICE50+ model. The source projections aggregate cyclonic wind, coastal, fluvial, pluvial flooding, and marine extremes (extreme waves, overtopping, heat, non-cyclonic wind) with concurrent growth in port exposure. Indexing these to GMST provides a single, transparent driver consistent with other sectors while retaining the net, scenario-path dynamics. In future work, these regressions could be re-specified with hazard-explicit drivers from the climate module (e.g., sea-level rise and extreme sea levels, storm surge, wave climate, tropical-cyclone proxies, compound flood metrics) coupled to an exposure module for port assets and throughput. It is important to note that not all disruptions reflected by EVaR will result in trade losses; many may simply cause delays. Consequently, only a portion of the risk to trade flows is expected to translate into economic losses. Estimating this proportion falls outside the scope of this study; therefore, we assume that EVaR is directly translated into losses.

For future GDP calculations, we factor in port-related EVaR using the damage function coefficients from Figure S14 C. These losses are then deducted from the consumption absent climate change $\tilde{C}$, as follows:

$$C_{c,t} = \tilde{C}_{c,t} \cdot \left(1 + \lambda_{c,t}^{ports} \Delta T_t \right) \quad \text{Eq. P1}$$

As mentioned, by including the trade risk in our EVaR estimates, we potentially overestimate the economic losses countries may experience as a result of climate-related disruptions at ports. On the other hand, some economic losses are not captured in our estimates. First, port-related disruptions could have cascading effects on the broader economy due to supply-chain disruptions, as highlighted in Verschuur et al²⁹. We refrain from including those cascading effects. Second, even disruptions that do not result in trade losses will still cause economic losses through costly delays and rerouting. Third, ports may provide benefits beyond their market value, such as cultural significance; however, in the RICE50+ model, they are considered only for their role in facilitating global ocean-based trade. As such, our EVaR is a compromise between overestimating some of the economic loss terms and underestimating the total EVaR by excluding some loss components.

In other words, in our model, ports are included only in terms of how they affect market output. When climate impacts diminish port functionality, it leads to a decrease in market output, reducing the consumption component of the utility function.

Fisheries and Mariculture

Overview

Fisheries and mariculture contribute to both market and non-market values, which are estimated separately using distinct methodologies. Market values are assessed through changes in fisheries profits under climate scenarios, while non-market use values are derived from the health benefits of seafood-derived nutrients. Each pathway follows a tailored workflow before converging into the Blue RICE50+ model.

To estimate the market value losses from climate impacts on fisheries, we use country-level projections of fisheries profits under different climate scenarios. The projections used from Free et al. assume optimal management, isolating the effects of climate change by comparing profits under RCP4.5, RCP6.0, and RCP8.5 against the low-warming baseline of RCP2.6. These differences, normalized by future GDP under SSP2, reflect the pure climate-driven impacts on fisheries profitability. We then fit country-specific damage functions linking profit declines to increases in global mean surface temperature. The resulting market damages are integrated into the Blue RICE50+ model. The full workflow is illustrated in Figure S15.

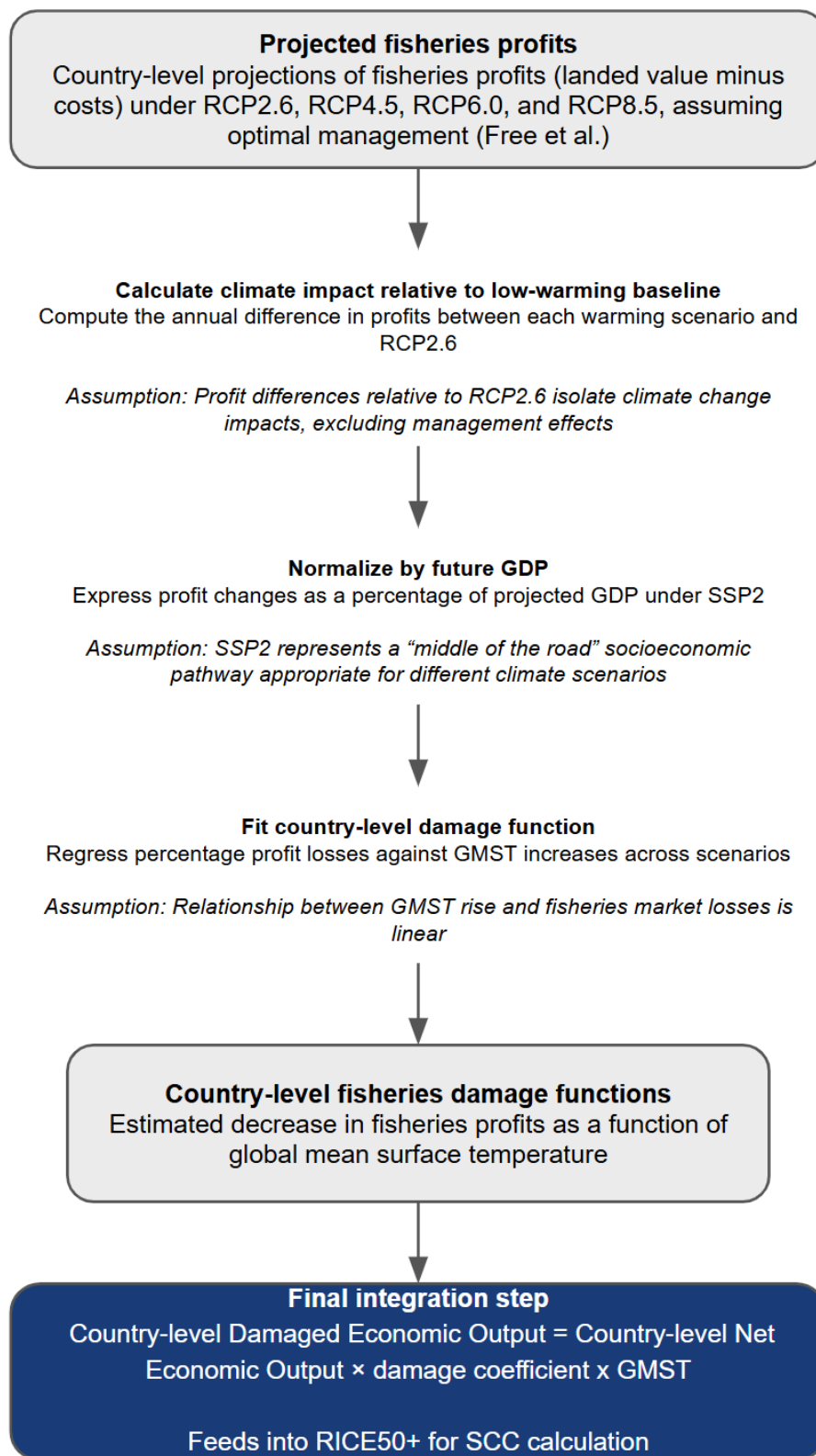


Figure S15. Workflow for estimating market value losses from fisheries under climate change.

This diagram outlines the steps used to estimate country-level fisheries market damages. Country-specific projections of fisheries profits under various RCP scenarios are compared against a

low-warming RCP2.6 baseline to isolate climate impacts. Profit changes are normalized by future GDP under SSP2 and regressed against projected global mean surface temperature increases to develop country-level damage functions. These damage functions are incorporated into the Blue RICE50+ model to assess the economic consequences of climate-driven changes in fisheries profitability.

To estimate the non-market use value losses from climate impacts on fisheries and mariculture, we focus on the nutritional role of seafood in supporting human health. Building on projections from Cheung et al., we first quantify how warming reduces the availability of key nutrients—calcium, omega-3, protein, and iron—at the country level. We then link these declines to increases in disease risk, based on a systematic review of epidemiological meta-analyses relating nutrient intake to mortality outcomes. Combining these effects with each country’s reliance on seafood for nutrient intake and baseline mortality rates, we estimate the additional deaths attributable to warming-driven nutrient losses. Finally, these excess deaths are monetized using a globally uniform value of a statistical life (VSL), adjusted over time with global income growth. The overall workflow is summarized in Figure S16.

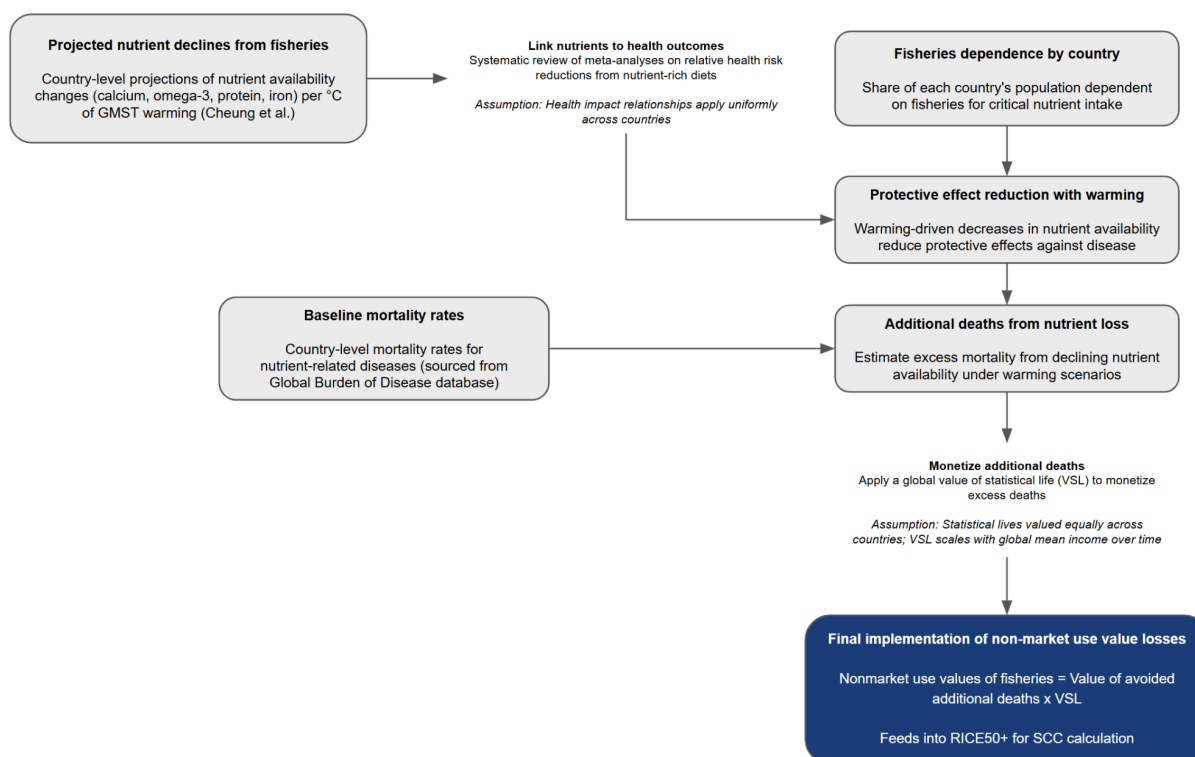


Figure F16. Workflow for estimating non-market use value losses from fisheries and mariculture due to nutrient declines.

This diagram illustrates the steps used to estimate welfare losses from reduced seafood-derived nutrition under climate change. Country-level projections of nutrient declines (calcium, omega-3, protein, and iron) are linked to increased disease risks based on a systematic review of epidemiological meta-analyses. Population-level fisheries dependence and baseline mortality rates are combined to estimate excess deaths attributable to nutrient losses. These additional deaths are monetized using a globally uniform value of a statistical life (VSL) that scales with global income over time. The resulting non-market use

value losses are integrated into the Blue RICE50+ model to assess welfare impacts of fisheries-related nutritional decline under future warming scenarios.

Market values of fisheries

Rising ocean temperatures are altering fish distributions⁶¹ and reducing the maximum sustainable yield⁶², which in turn impacts the catch sizes and profitability of national fisheries⁶³. Recent research by Free et al.⁵² have quantified the impact of climate change and management practices on a large portion of global fisheries, representing 60% of the total catch. Using these projections at the national level, we examine the specific impact of climate change on the profits of fisheries under their hypothetical "Full Adaptation" scenario. This scenario assumes that fishers adjust their harvest practices to maintain the maximum sustainable yield in the face of climate-induced productivity shifts, and continue to do so even as fish populations move across international boundaries. Free et al. derived profits by subtracting the costs of fishing from the total catch multiplied by the ex-vessel price, obtained from a global reconstruction study⁶⁴. Extended Data Table 6 presents the profits per kilogram and total catch in 2025 for the 780 species accounted for in the study.

We illustrate our country-specific workflow using Mexico's fisheries as an example. The top part of Figure S17 A illustrates the profit trajectories from Mexico's fisheries across the century under various climate change scenarios. In all scenarios, short-term profits fall as management reduces fishing mortality. After this dip, profits remain stable under the least severe climate change scenario (RCP 2.6, green). Initially, profits rise under RCP 4.5 (blue) and 8.5 (red) as a result of range expansions that increase productivity and result in the movement of new stocks into the country's Exclusive Economic Zone (EEZ). Profits fall precipitously under the most severe climate change scenario (8.5, red) and settle at the RCP 2.6 levels for scenario RCP 4.5 (blue). Profits fall consistently under RCP 6.0 (yellow).

The bottom part of Figure S17 A displays these profits as a percentage of GDP over the century, based on GDP projections from SSP2 (which roughly involves the climate forcings equivalent to those used in RCP 4.5) across all RCP scenarios. This approach isolates the impact of climate change on profits, excluding the influence of exogenous GDP trajectories. Currently, the lack of integration between SSPs and scenarios for fishing technologies and fleets leads to profit projections that assume a static external economy, with the number and type of fishing vessels fixed at 2015 levels for each country. This assumption may be revisited in future work as the community developing fishing projection models begins to integrate the SSPs with the Oceanic System Pathways⁶⁵.

To assess the impacts of climate change on fisheries profits, we will reframe the analysis from tracking profits as a percentage of GDP over time to examining annual profits and temperature deviations from the minimal climate change scenario (we use RCP 2.6 as the baseline due to a lack of a future profit trajectory under a scenario without climate change). This means that for each year, we will compare the deviation in profits and temperatures under scenarios RCP 4.5,

6.0, and 8.5 from those under RCP 2.6, called $D_t^{fisheries} = Profits_{t,RCP \neq 2.6} - Profits_{t,RCP2.6}$. This method of re-mapping is depicted in Figure S17 B. Here, we see that as the global mean surface temperature increases from the baseline scenario, the share of GDP attributable to fisheries in Mexico decreases. The larger circles in Figure S17 B highlight the differences in values from the year 2075, which are highlighted in Figure S17 A (bottom), to illustrate how this coordinate transformation is applied.

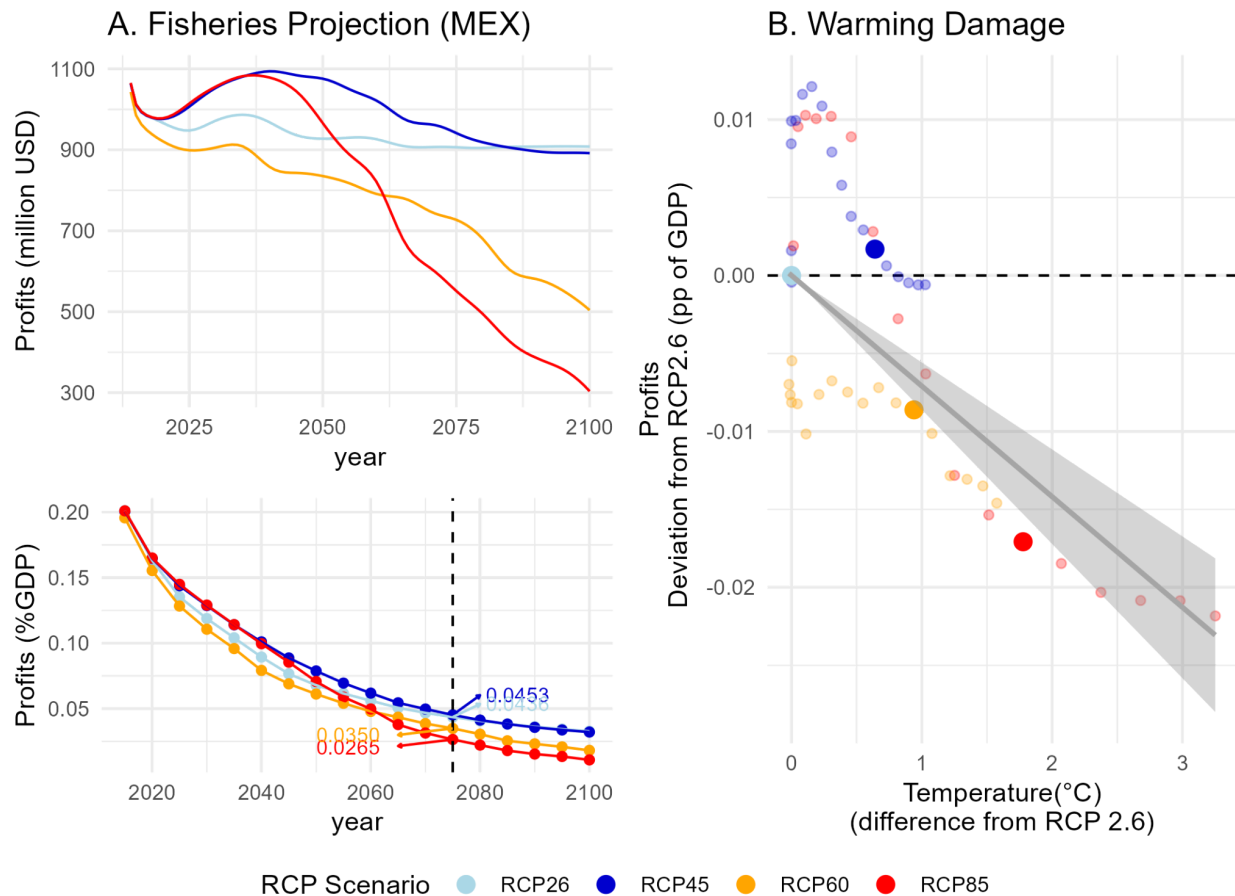


Figure S17. A: Profit trajectories from Mexican fisheries under different warming scenarios, top: in million USD, bottom: as percentage of GDP. B: Deviation of profit trajectories from scenario RCP2.6, for example, for a 2°C warming over RCP2.6 the GDP level is reduced by 0.02%. .

Once we have determined the year-to-year differences in profits and temperature deviations from RCP 2.6 under various scenarios, we can derive a damage function. This function is a linear regression model that fits the points in Figure S17 B, illustrating the average profit loss associated with each degree of warming. We compute the damage function coefficients for every country globally, refining them to reflect the broader economic impact of fisheries:

$$D_t^{fisheries} = \lambda_{1,c}^{fisheries} \Delta T_{t \text{ from RCP2.6}} + \epsilon_t \quad \forall c \quad \text{Eq. F1}$$

The damage $D_t^{fisheries}$ also incorporates country-level economic multipliers, as estimated by Dyck and Sumalia (2010)⁵³, which are presented as regional average values in Table S4. These multipliers quantify the downstream effects of an additional dollar of landed fishery value on other economic sectors and incomes. Essentially, the derived damage functions measure the impact of climate change not only on the initial market sale profits of fisheries but also on a fraction of the wider economy.

Table S4. Economic and Income multipliers for fisheries. Obtained from Dyck and Sumalia⁵³.

Region	Economic Multiplier	Income Multiplier
Africa	2.59	0.62
Asia	2.67	0.71
Europe	3.12	0.76
Latin America	2.05	0.56
North America	3.52	1.22
Oceania	3.27	0.73

The world map in Figure S17 shows the damage function coefficients $\lambda_{1,c}^{fisheries}$, revealing that the majority of countries experience negative impacts from climate change on their fisheries, with only a few poleward countries, such as Canada, Russia, Iceland, and Norway, showing positive impacts (Extended Data Table 7).

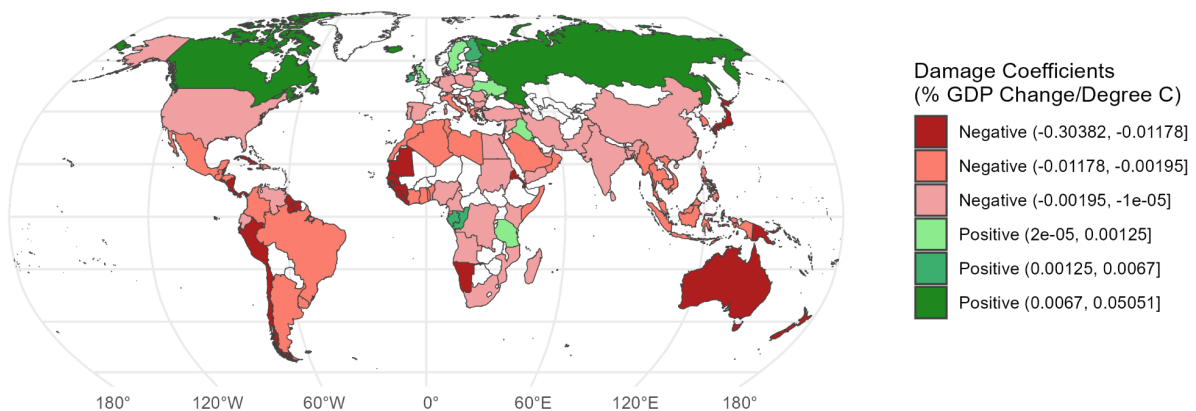


Figure S18. Climate change impacts on fisheries profits as a percentage of GDP per degree of warming. Global map of country-level damage function coefficients representing the change in fisheries-related profits, as a share of GDP, for each additional degree of global mean temperature increase. Most countries experience negative impacts, while a few higher-latitude nations show potential gains due to poleward shifts in fish stocks.

We use these coefficients to include the fisheries-related losses in the RICE50+ model. We assume that the current economic output at the country-level in RICE50+'s economic module already includes all the current fisheries-related values. For future GDP calculations, we factor in fisheries-related impacts using the damage function coefficients from Figure S18. These losses are then deducted from the market consumption ($\tilde{C}$) as follows:

$$C_{c,t} = \tilde{C}_{c,t} \cdot \left(1 + \lambda_{c,t}^{fisheries} \Delta T_t\right) \quad \text{Eq. F1}$$

It's crucial to understand that these coefficients should be seen as a lower bound estimate as they only account for 60% of fisheries worldwide (so losses could be greater when accounting for missing fisheries) and we are using the "Full Adaptation" scenario, which is a best-case adaptation scenario. Although we employ a multiplier factor to illustrate the impact of fisheries damages on more sectors, this estimate still represents a lower bound because it only includes a limited number of sectors. In addition, other values overlooked in this fisheries analysis include non-market aspects such as the cultural significance of ancestral fishing techniques.

Health Benefits from Fisheries and Mariculture

Rationale and Analysis

Fisheries and marine aquaculture (mariculture) are important sources of nutrients for many populations around the world. As shown by Cheung et. al (2024)³¹ these nutrient sources will decline globally, impacting disproportionately more on lower-income countries (Figure S19). In their analysis, Cheung et al. reconstructed historical fish catches and mariculture over time at the country level using databases from the Sea Around Us project⁶⁶. The catches from these sources were then translated into the content of nutrients such as calcium, iron, protein, and omega-3 fatty acids based on a database of published nutrient composition studies and a trait-based Bayesian model⁶⁷. Finally, the nutrients available from seafood consumption were scaled based on three databases of edible portions for different classes of seafood.^{68,69}

Future nutrient availability for human consumption was calculated by Cheung et al. using the same approach as before but under projections of potential catches of marine fishes and invertebrates using model output from the Dynamic Bioclimate Envelope Model (DBEM), which is a process-based model that simulates biological and ecological processes of marine ecosystems^{30,70}. Changes in mariculture were obtained from the Global Mariculture Production

Model, which uses future climatic and socioeconomic conditions to project mariculture production potential.^{71,72} Finally, a damage function was estimated to regress the nutrient availability percent change ΔA on the change in mean global surface temperature (ΔT):

$$\Delta A_{n,r,t} = \beta_{n,r} \Delta T_t$$

In this equation, n denotes the various nutrients analyzed, including calcium, iron, protein, and omega-3 fatty acids, while r categorizes the country income levels as analyzed by Cheung et al., distinguishing between low-income and high-income groups.

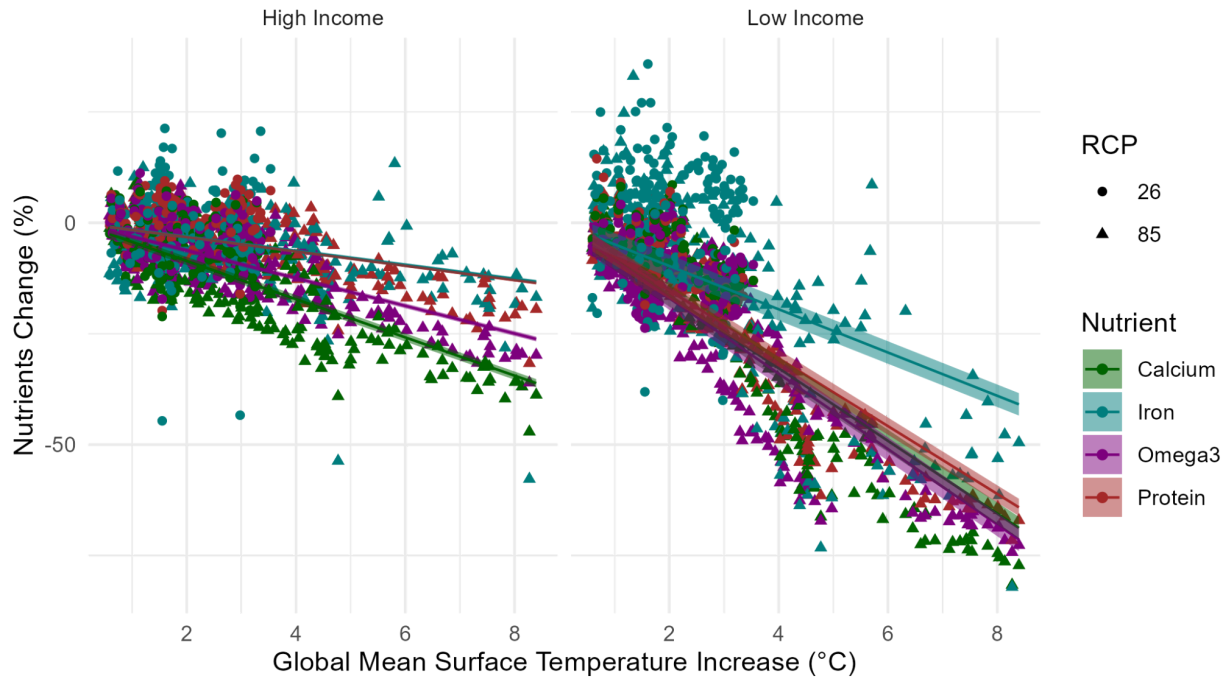


Figure S19. Projected changes in seafood-derived nutrient availability by income group. Projected percent change in the availability of key nutrients (calcium, iron, protein, and omega-3) from fisheries and mariculture as a function of global mean surface temperature increase above pre-industrial levels. Results are shown separately for high-income and low-income countries, based on estimates from Cheung et al. (2024).

Building on the analysis of climate-induced changes in nutrient availability, we established the relationship between nutrient intake and health outcomes using established meta-analyses from the epidemiological literature. We performed a systematic review, extracting meta-analysis studies from the Web of Science that provided quantifiable risk estimates concerning nutrient intake, including their associated uncertainties. The search string used was:

(("relative risk") AND ("standard error" OR "SE" OR "confidence interval" OR "CI" OR "p-value" OR "p value") AND "meta-analysis") AND (("omega-3" OR "omega 3" OR "n-3 fatty acid*" OR "EPA" OR "DHA" OR "fish oil") OR ("calcium" OR "Ca" OR "dietary calcium") OR ("iron" OR "Fe" OR "dietary iron" OR "heme iron" OR "non-heme iron") OR ("protein"))

We continued this review until reaching a point of saturation, where the inclusion of additional studies no longer contributed significant new insights. This effort yielded a comprehensive database that delineates the extent to which diets rich in specific nutrients can reduce the risk of particular health conditions. Table S5 presents a simplified version of this database, while a complete compilation of all reviewed studies, including those not utilized for various reasons, is available in Extended Data Table 8.

Table S5. Health outcomes associated with nutrient intakes.

Nutrient (n)	Health outcome (h)	Reduction of Relative Risk (ρ)	95% CI
Omega3	Cardiovascular diseases	0.07	(0.02,0.12)
Omega3	Non-fatal myocardial infarction	0.13	(0.07,0.19)
Omega3	Coronary heart disease event	0.09	(0.04,0.13)
Omega3	Major adverse cardiovascular event	0.05	(0.02,0.08)
Omega3	Revascularization	0.09	(0.05,0.13)
Omega3	Preeclampsia	0.18	(0.3,0.03)
Omega3	Neonatal preterm birth	0.49	(0.07,0.6)
Omega3	Colon and rectum cancer	0.12	(0,0.22)
Omega3	Asthma	0.38	(0.09,0.66)
Omega3	Endometrial Cancer	0.05	(0.00,0.09)
Omega3	Breast cancer	0.06	(0.03,0.09)
Omega3	Liver cancer	0.18	(0.04,0.3)
Calcium	All causes	0.09	(0.06,0.12)
Iron	Gestational diabetes mellitus	0.08	(0.05,0.13)
Iron	Cardiovascular diseases	0.28	(0.12,0.41)

The estimates of the reduction of relative risk (ρ) reported in the meta-analysis provide estimates on how much the risk of specific health outcomes (h) is decreased for a given population that received a treatment compared with a population that didn't received it. In this case, the treatment refers to a population that had higher nutrient intake. We assume that current population globally has a full nutrient intake and therefore the present risk (R) associated with health outcomes (h) is lower than what they would have without nutrients intake.

$$R_{h,t,n,r} = (1 + \beta_{n,r} \Delta T_t) \cdot \rho_{h,n}$$

For example, a full intake of omega-3 fatty acids reduces risk of cardiovascular diseases by 7% relative to a diet poor of omega-3 fatty acids. Assuming that such risk reduction decreases proportionally to nutrient availability, we can project the changes in risk reduction under different warming scenarios as shown in Figure S20.

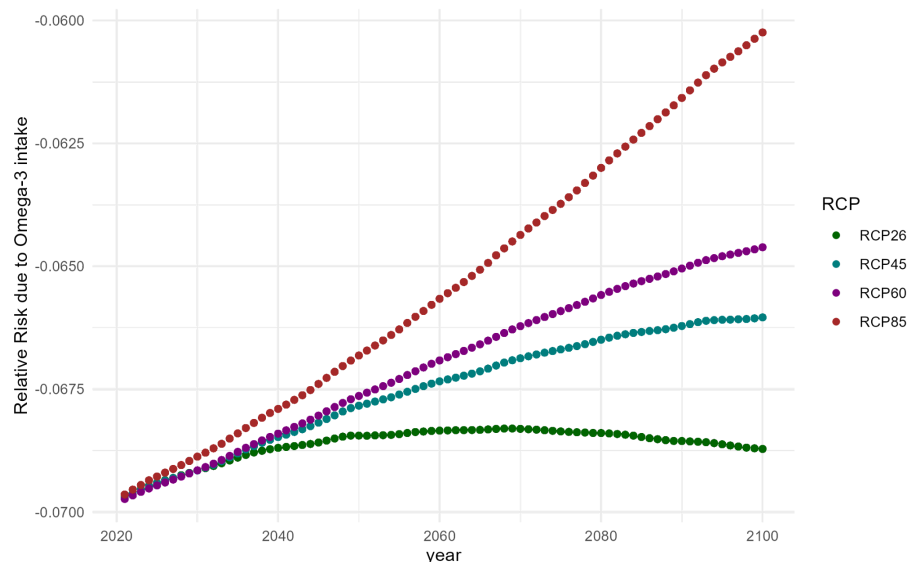


Figure S20. Projected reduction in cardiovascular disease protection from omega-3 fatty acids availability in the future. Estimated decline in the protective effect of omega-3 intake on cardiovascular disease risk under future warming scenarios. The figure shows how climate-driven reductions in omega-3 availability from seafood are expected to weaken the associated health benefits, based on relative risk reductions reported in epidemiological meta-analyses.

We use the change in risk to estimate mortality change using the Global Disease Burden database⁷³, assuming that mortality will change proportionally to the risk increase, modulated by two parameters: percentage of population in each country dependent on fisheries (μ_c) as obtained by Cheung et al., and fraction of nutrient intake that cannot be made up by consuming other foods. Although local studies (e.g., Heilpern et al. 2021 in the Peruvian Amazon⁷⁴) show that dietary substitution is possible, they also indicate that substitutes (e.g., farmed animals) may not provide equivalent levels of essential nutrients such as iron and fatty acids. In the absence of global estimates, we therefore adopt a judgment-based parameter of $\eta = 0.05$ that indicates the fraction of nutrients that cannot be substituted for. This deliberately conservative choice reflects the assumption that the vast majority of lost nutrient intake can be compensated, while ensuring that some residual damages remain to capture the imperfect substitutability highlighted in the empirical literature. The mortality change is given by:

$$\Delta MR_{h,t,n,r,c} = \Delta R_{h,t,n,r} \cdot \eta \cdot \mu_c \cdot MR_{h,t0,c}$$

Where h indicates the health condition, t is the year, n is the associated nutrient, r is the income group, and c is the country of interest. Figure S21 shows the evolution of all mortality associated with nutrient-dependent health conditions. As shown in the graph, the largest contributor to additional deaths comes from calcium availability decrease, this is because the meta-analysis found for calcium intake took into account the reduction in relative risk for all types of health conditions, while the meta-analysis found for iron and omega-3 were more focused on specific

health conditions. The meta-analysis that we found for risk reduction related to protein intake are substantially more detailed by type of food and available substitutes, which require a more nuanced approach for the scope of this paper and therefore we are not able to include them in this estimate.

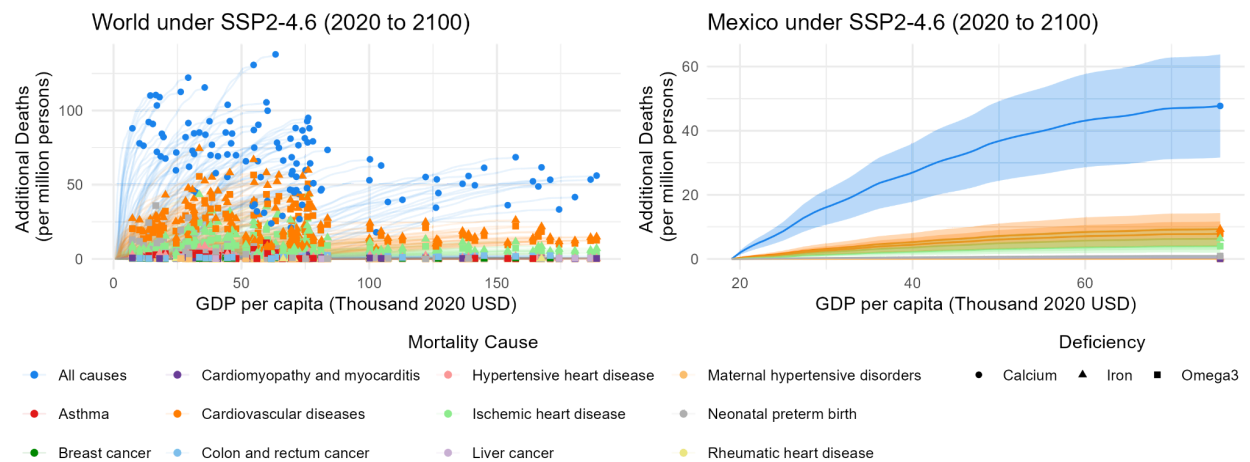


Figure S21. Projected additional deaths from nutrient deficiencies by cause and nutrient. Country-level projections of excess mortality associated with declining availability of calcium, iron, and omega-3 due to climate impacts on fisheries and mariculture. The largest contributor to additional deaths is calcium deficiency, driven by broader health risk coverage in the underlying meta-analysis. Estimates for protein-related mortality are excluded due to data limitations on substitution and specificity.

Aggregating additional deaths by all types causes we obtain a country-level projection of additional premature deaths (Figure S22). Which is sensitive to the climate condition via the reduction of nutrient availability and to the socioeconomic scenario via the total population.

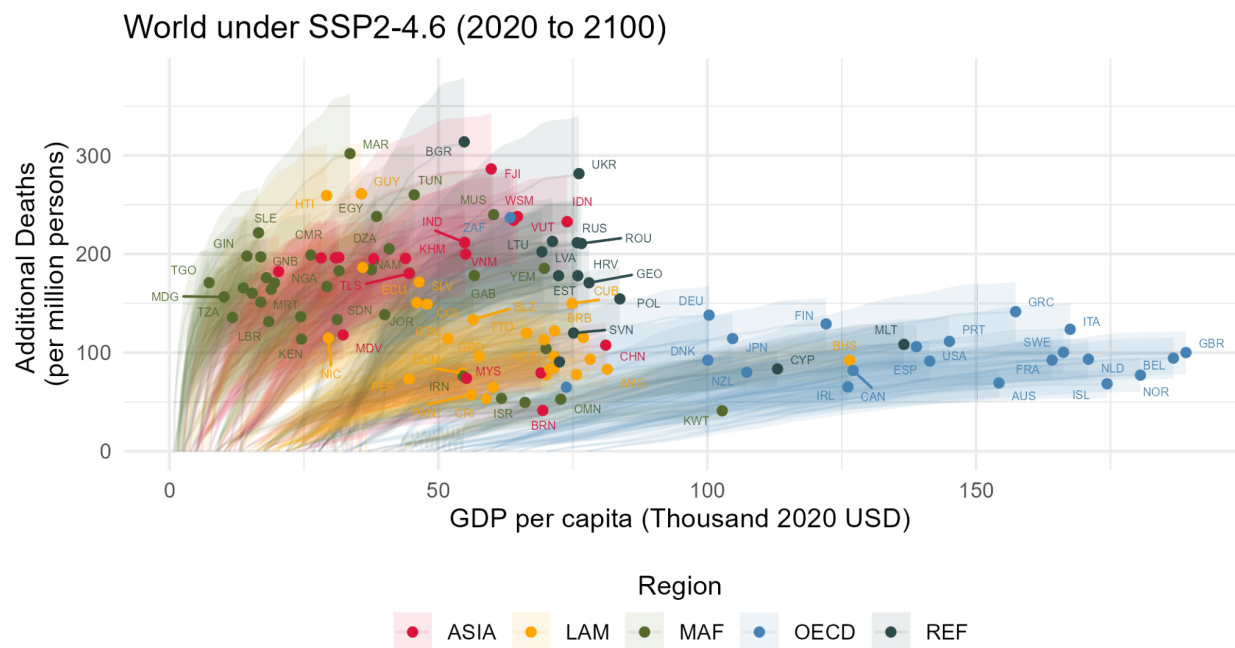


Figure S22. Projected additional deaths due to seafood-derived nutrient deficiencies, by country. Country-level estimates of excess premature deaths attributable to climate-driven reductions in seafood-derived nutrients (calcium, iron, omega-3).

Conversely, the additional deaths could be seen as a reduction in avoided premature deaths due to seafood consumption, as shown in Figure S23.

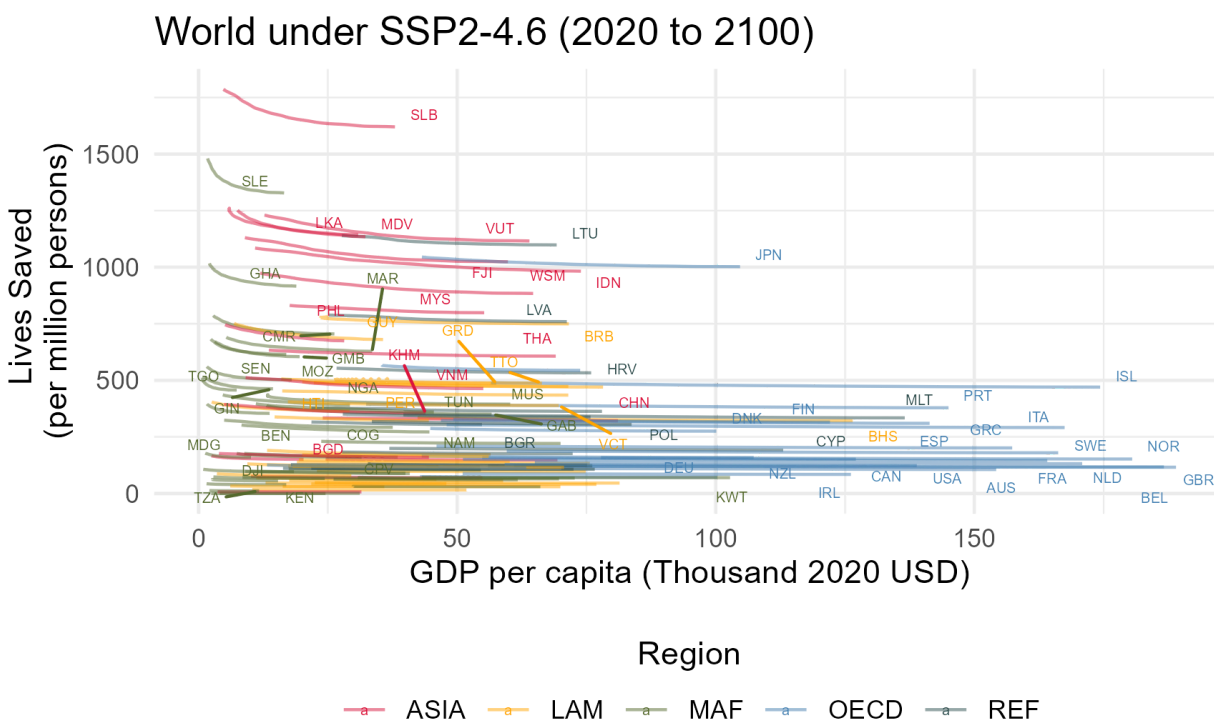


Figure S23. Decline in avoided premature deaths due to decreasing seafood-derived nutrient availability. Projected reduction in the number of premature deaths avoided through seafood consumption from 2020 to 2100, by country. This reflects the diminishing health benefits of nutrients such as calcium, iron, and omega-3 as their availability declines under climate change.

Our nutrition-mediated channel (via fisheries/mariculture losses) implies a change in annual mortality risk of 1.79×10^{-5} per person at 1°C . For reference, the 2023 EPA health modules imply $\approx 1.82 \times 10^{-4}$ in DSCIM under SSP3 with no explicit adaptation and $\approx 3.748 \times 10^{-3}$ in GIVE under the RFF-SP scenarios, both evaluated at $\sim 1^\circ\text{C}$. While socioeconomic pathways and the treatment of adaptation differ across these models and can materially affect levels, our estimate is at least an order of magnitude lower than GIVE and DSCIM mortality estimates. As a comparator from the agriculture–nutrition literature, Springmann et al.⁷⁵ model climate-driven agricultural yield shocks, their impacts on diets and body weight, and the downstream effects on cause-specific mortality. Their 2050 projection (averaged across RCPs) implies an increase of $\sim 0.8 \times 10^{-5}$ in per-person annual mortality risk. This is a net 2050 quantity that includes putative benefits from lower body mass index (BMI) among overweight populations—a component that remains debated^{76–78}. Given these differences in timing of the damage (at 1°C vs. 2050), and mechanism (diet change with BMI offsets vs. marine micronutrient pathways), the Springmann estimate is broadly commensurate with our fisheries and mariculture values.

In addition, we tested the sensitivity of our result to national undernourishment prevalence, which shifted an illustrative 1°C mortality increment from $\sim 1.7 \times 10^{-5}$ to $\sim 1.5 \times 10^{-5}$, but did not adopt it here because national estimates are likely overestimates of coastal-only nutrient intake, especially for our nutrients of interest.^{79,80} In addition, today's nutrition burden is already

reflected in baseline cause-specific mortality rate, and SSP-consistent, coastal-level projections are not yet available; we flag this for future work using coastal nutrition microdata and potentially including this in efforts to replicate SSPs for ocean-relevant metrics, like the Oceanic System Pathways.⁸¹

Integration to RICE50+

To integrate these seafood-dependent health benefits in RICE50+'s utility function as a non-market use value, we aggregate the different pieces of the analysis above into a single equation that estimates the monetized health benefits from seafood nutrition:

$$HealthBenefits_{c,t} = (1 + \beta_c T_t) \cdot TAME_c \cdot Pop_{c,t} \cdot \mu_c \cdot \eta \cdot VSL_t$$

where $TAME_c$ is the total avoided mortality effect of country c , defined as the sum of the baseline mortality rate (MR) associated with health condition h , times the protective effect (ρ) that each nutrient n has on such condition. It is a per capita measure that quantifies the effectiveness of seafood-derived nutrients in reducing the risk of death from various health conditions within that country.:

$$TAME_c = \sum_{(n,h) \in M} \rho_{h,n} \cdot MR_{c,h,2020}$$

M is the set of specific pairs (n, h) for which we have values in our meta-analysis dataset. The coefficient β_c is the weighted average of the coefficients $\beta_{n,r}$ from Cheung et al., using as weights the $TAME_c$. This weighted coefficient β_c reflects the sensitivity of the total protective effect from nutrients to the temperature, weighted by the nutrients effect on health outcomes in a specific country. The vector μ_c is the dependence of each country on fisheries, ranging from 0 to 1, and the parameter η , assumed to be 0.05, is the capacity of each country to overcome the fisheries-based nutrient deficiency with other sources of food. The economic valuation parameter VSL_t (the willingness to pay to avoid a very small risk of death) is assumed to be the same across countries and it changes across time, following recommendations by Bressler and Heal⁸². In particular, VSL grows with global population-weighted mean GDP per capita:

$$VSL_t = \widehat{VSL} \cdot \frac{Global\ GDPpc_t}{Global\ GDPpc_{2020}}$$

For the year 2020, we adopt the VSL estimate proposed by the U.S. Environmental Protection Agency's guidelines for preparing economic analyses⁵¹, which fits a Weibull distribution to 26 studies with a central value of \$7.4 million (US-\$[2006]), and a standard deviation of \$4.7 million (US-\$[2006]). To better reflect average global income levels, we rescale this U.S. VSL by the

ratio of global GDP per capita to U.S. GDP per capita, yielding an adjusted value of \$2.15 million in 2020.

To reflect equity concerns in global climate policy, we apply this value uniformly to all countries rather than adjusting it by national income levels. This avoids assigning lower value to lives in lower-income countries—a criticism of past approaches. For instance, previous IPCC assessments that relied on regional VSLs based on local incomes drew public criticism for implying that a life lost in a low-income country was worth significantly less than one in a high-income country—raising ethical concerns and confusion around whose welfare was prioritized in global damage estimates^{83–86}.

We scale the VSL over time using an income elasticity of 1, consistent with benefit transfer guidelines and empirical studies that show willingness to pay for mortality risk tends to rise proportionally with income. While the empirical literature includes a range of elasticity estimates, many policy institutions and academic sources recommend using an elasticity of 1 for international applications, as it balances theoretical consistency with empirical robustness. For example, Viscusi and Masterman⁸⁷ recommend a central value of 1.0 when transferring VSLs across income contexts in global analyses. This assumption implies that VSL grows proportionally with income—simplifying the analysis while avoiding the risk of disproportionately inflating mortality damages in high-income countries, which could otherwise result from using higher elasticity values. Thus, this choice also helps prevent ethical and interpretative challenges that have emerged in past SCC applications.

While we do not implement a fully welfare-weighted approach that adjusts VSL values by region, we incorporate mortality damages into the broader economic framework of the RICE50+ model. In this model, all damages—including those from mortality, changes in ecosystem services, and market losses—are aggregated into a single social welfare function. The model applies a globally uniform structure of Ramsey discounting: the same pure rate of time preference and elasticity of marginal utility are used across regions. Country-level differences in how future damages are valued emerge naturally based on differences in economic growth. In this way, although the VSL is uniform across space, the relative weight of mortality impacts over time and across countries is determined endogenously by the model. Because all monetized losses—including those from health—are funneled through a shared welfare function, our approach preserves consistency with utility-based discounting while reflecting a deliberate equity stance in valuing human life.

Monte Carlo analysis

We compute the distribution of the blue social cost of carbon and its components sampling the input parameters and computing the SCC in 10,000 simulations. The parameters are drawn with Latin Hypercube sampling from the distributions described in Table S6. This sampling scheme offers improved efficiency over random sampling. The distribution of the SCC is plotted in the lower panel of Figure 3.

We further explore the *sensitivity* of the blue social cost of carbon to input parameters over a second set of 10,000 simulations. The parameters are also drawn with Latin Hypercube sampling from the distributions described in Table S6. However, we intentionally restrict the sensitivity analysis to elements that are specific to the formulation of the ocean-related SCC and exclude inputs whose importance for the SCC has already been extensively discussed, such as the transient climate response to cumulative emissions and the discount rate.

We summarize the sensitivity of the SCC to parameter variation via the PAWN index⁸⁸. The index uses cumulative distribution functions (CDFs) to compare the unconditional distribution of SCC (when all parameters are varied) with conditional distributions (when one parameter is fixed at a time). The difference between the unconditional and conditional CDFs is assessed using the Kolmogorov–Smirnov test, which measures the maximum distance between these distributions. The final PAWN index for a parameter is calculated by taking a summary statistic (like mean, median, or maximum) over all the Kolmogorov–Smirnov values computed between the unconditional and conditional distributions. We report here the mean of such values. The scale of the index ranges from 0, meaning no dependence of the SCC on the parameter, to 1, indicating full dependence.

Figure S24 illustrates the PAWN index values and 95% confidence intervals obtained with bootstrapping. The value of the ocean-related social cost of carbon is most sensitive to the elasticity of substitution of the three forms of blue capital (θ). This parameter controls how much the utility loss from a unit reduction of a form of blue capital can or cannot be compensated by increasing other forms of blue capital or consumption. Higher values of θ represent higher substitutability and imply a lower social cost of carbon. The non-normative input parameters that yield the largest influence on the SCC are those that regulate the value of its largest components, namely those stemming from the non-use value of corals and the non-market use values of fisheries. In particular, the non-use value per square kilometer of corals by Brander et al. (2024), the fraction of nutrients from fisheries that cannot be substituted (η), assumed to be 5%, and the Value of Statistical Life.

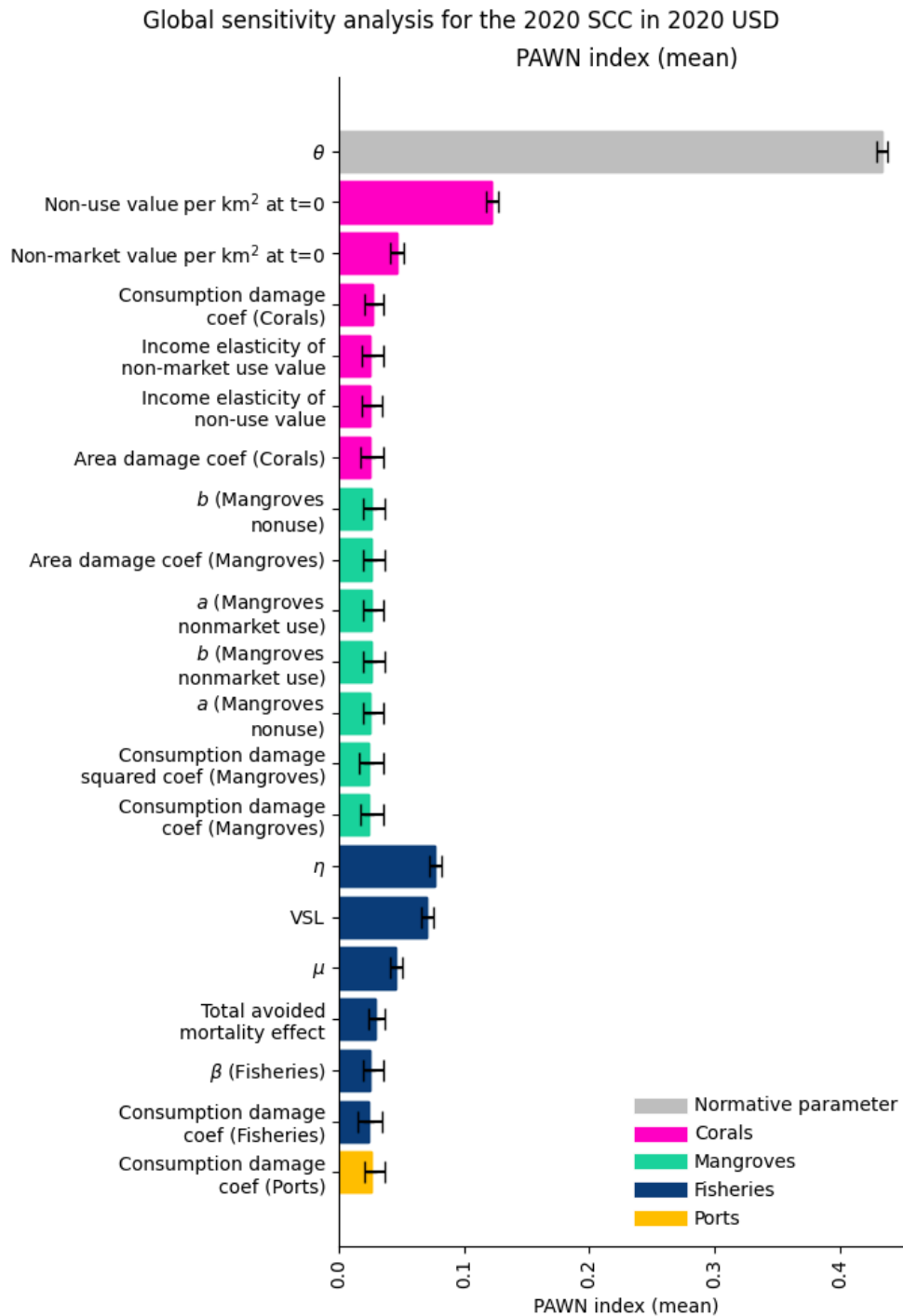


Figure S24. Sensitivity of the Blue Social Cost of Carbon to key input parameters (PAWN index). Mean PAWN index values and 95% confidence intervals showing the influence of each input parameter on the blue SCC, based on global sensitivity analysis with bootstrapping. The elasticity of substitution between blue capital components (θ) is the most influential parameter. Other important drivers include the non-use value of corals, the fraction of non-substitutable nutrients from fisheries, and the Value of Statistical Life (VSL).

Table S6. Parameter values and distribution.

	Parameter	Value	Distribution	Category	Valuation type	Source
% of population dependent on fisheries	μ	varies by country	U(-0.25%, 0.25%)	Fisheries	Non-market use	Cheung et al.
% of nutrient intake from fisheries that cannot be substituted	η	0.05	U(0.01, 0.2)	Fisheries	Non-market use	
Starting non-market use values per area	$p^{\text{nonmarket}}$	varies by country	Normal(μ, σ)	Corals, Mangroves	Non-market use	Brander et al.
Starting non-use values per area	p^{nonuse}	varies by country	Normal(μ, σ)	Corals, Mangroves	Non-use	Brander et al.
Coefficient for nonmarket use values	$a^{\text{nonmarket}}$	varies by country	Normal(μ, σ)	Mangroves	Non-market use	Estimated
Coefficient for nonmarket use values	$b^{\text{nonmarket}}$	varies by country	Normal(μ, σ)	Mangroves	Non-market use	Estimated
Coefficient for nonuse values	a^{nonuse}	varies by country	Normal(μ, σ)	Mangroves	Non-use	Estimated
Coefficient for nonuse values	b^{nonuse}	varies by country	Normal(μ, σ)	Mangroves	Non-use	Estimated
Area damage coefficients	δ	varies by country	Normal(μ, σ)	Corals, Mangroves	Non-market use, Non-use	Estimated
Consumption damage coefficients	λ	varies by country	Normal(μ, σ)	All	Market	Estimated
Total avoided mortality effect	TAME	varies by country	Normal(μ, σ)	Fisheries	Non-market use	Derived from multiple inputs

Coef for mortality-temperature relation	β	varies by country	Normal(μ, σ)	Fisheries	Non-market use	
VSL	VSL	7.4	Positive Normal(7.4, 4.7)	Fisheries	Non-market use	EPA
Utility function parameter	θ	0.21	Positive Normal(0.21, 0.09)	All		-
Utility function parameter	$\gamma^{\text{nonmarket}}$	0.222	Positive Normal(0.79, 0.09)	Corals	Non-market use	-
Utility function parameter	γ^{nonuse}	0.243		Corals	Non-use	-
TCRE	TCRE	1.6	LogNormal(0.5, 0.43)	All		
Pure rate of social time preference		X	U(0.01, 0.2)	All		
Elasticity of marginal utility of consumption		1.45				

Results for other SSPs

We use SSP2 as our central scenario, but explore alternative socio-economic futures under different Shared Socioeconomic Pathways (Figure S25). The blue SCC varies across these scenarios, ranging from approximately \$42 per ton of CO₂ under SSP5 (a fossil-fueled development pathway with high economic growth and reduced inequality) to \$87 per ton under SSP3 (a pathway of regional rivalry, high inequality, and elevated emissions). SSP2, our central case, yields an intermediate estimate of \$55, consistent with its “middle-of-the-road” characterization. SSP1 (sustainability-focused development) results in a slightly lower blue SCC of \$47, reflecting coordinated mitigation and reduced exposure to ocean-related damages. In contrast, SSP4 combines low to medium economic growth with persistent regional inequality, yielding a higher blue SCC of \$70. These results highlight how both emissions trajectories, income growth and inequality interact to shape the social cost of carbon from ocean-based impacts.

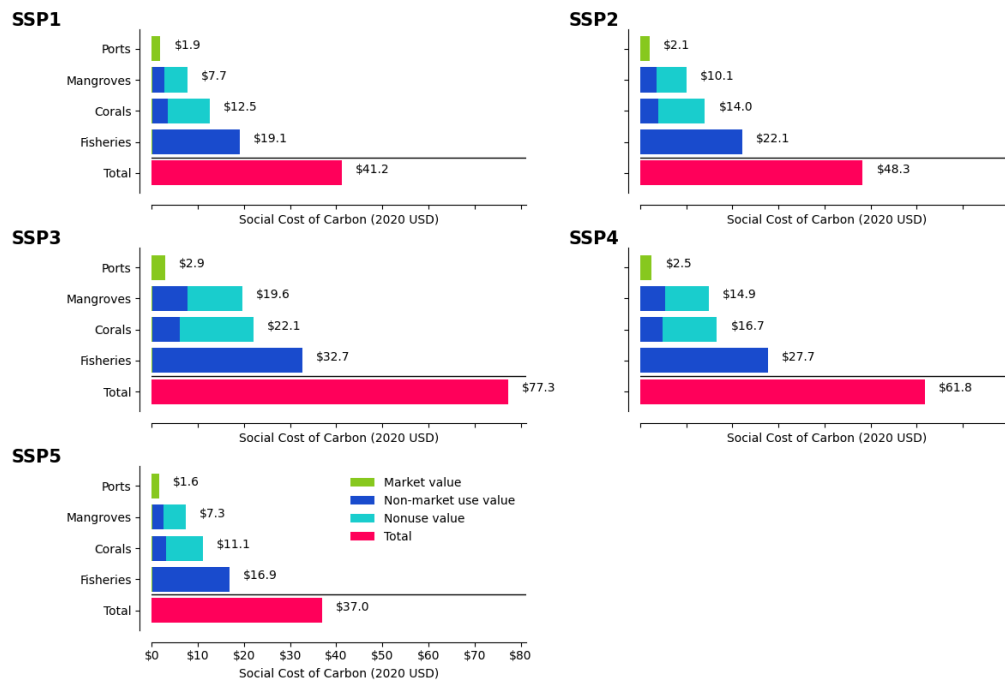


Figure S25. Blue Social Cost of Carbon under alternative Shared Socioeconomic Pathway (SSP) assumptions. The figure shows the estimated Blue SCC across five SSPs, reflecting different trajectories of socioeconomic development and associated emissions. Each SSP results in a different increase in global mean temperature by 2100 relative to pre-industrial levels. End-of-century GMT increases are as follows: SSP1: 2.60°C, SSP2: 3.58°C, SSP3: 4.18°C, SSP4: 3.82°C, SSP5: 3.95°C.

Results varying the pure rate of time preference and the elasticity of marginal utility of consumption

In our main analysis, we use a pure rate of time preference (prtp) of 1.5% and an elasticity of marginal utility of consumption (emuc) of 1.45, consistent with the default assumptions of the RICE50+ model. These values were inherited from the broader DICE/RICE modeling tradition, where they were originally chosen to calibrate the model's behavior to observed real interest rates and savings behavior. Our goal in using these defaults was to maintain comparability within the established RICE-based framework, particularly as our primary contribution lies in incorporating ocean-based damages into welfare analysis.

However, we acknowledge that more recent work, such as Rennert et al. (2022)¹⁶, has taken steps toward empirically estimating these parameters based on observed interest rates and economic growth data. While replicating this calibration for this analysis is out of the scope of this work, we take at face value the prtp and emuc pairs reported in Rennert et al. (2022) consistent with a near-term consumption discount rate of 1.5%, 2%, 2.5%, and 3%. The pairs

(0.01%, 1.02), (0.2%, 1.24), (0.5%, 1.42), and (0.8%, 1.57) yield 2020 Blue SCC estimates of \$305, \$168, \$96, and \$64 USD₂₀₂₀/tCO₂, respectively.

Comparison with Other Social Cost of Carbon Estimates

In this section, we provide further context on how our blue SCC compares to other published SCC estimates, particularly those used in U.S. regulatory settings or reported in recent academic literature, such as Rennert et al. (2022) and the U.S. Environmental Protection Agency (EPA) 2023 technical report.

Our study estimates the social cost of carbon associated with ocean-based welfare losses using a version of the RICE50+ integrated assessment model (IAM) that incorporates climate damages to four types of “blue capital” (corals, mangroves, fisheries and mariculture, and seaports). A defining feature of our approach is the use of a nested utility function that explicitly distinguishes between market, non-market use, and non-use values of nature. We compute welfare losses at the country level, convert those utility changes to monetary equivalents using each country’s marginal utility of consumption, and then aggregate monetary damages across countries (population-scaled).

After computing these country-level monetary damages, we aggregate them into a global total and apply a global Ramsey discount factor to compute the present value of future damages. The global discount factor depends on the pure rate of time preference ($\rho = 0.015$) and a time-varying component linked to the projected path of global per-capita consumption, consistent with the Ramsey framework.

This approach differs from commonly referenced SCC estimates (e.g., Rennert et al., 2022; EPA, 2023) in two key respects::

1. **Damage representation and welfare aggregation:**
We model damages inside a nested utility function (separating market, non-market use, and non-use values) and then obtain money-metric damages via marginal utility before global aggregation. By contrast, many estimates apply damages directly to output/consumption via reduced-form damage functions and aggregate in monetary units throughout.
2. **Scenario differences:**
Our estimates are based on a modified RICE50+ model that uses the shared socioeconomic pathways as scenarios for income and population growth. In contrast, Rennert et al. (2022) developed the GIVE (Greenhouse Gas Impact Value Estimator) model, which uses probabilistic socioeconomic projections, and applies a stochastic growth-linked discounting framework calibrated to recent real interest rate data.

As a result of these differences, direct comparisons between our blue SCC and estimates from EPA or Rennert et al. should be interpreted cautiously.

For the reasons outlined above, we focus our primary comparison on the difference between the Blue SCC and the SCC derived from the same model excluding ocean-related damages. Using the widely adopted damage function by Kalkuhl and Wenz⁸⁹—featured in the core version of RICE50+—the baseline SCC without blue damages is estimated at \$51 per ton of CO₂. Adding our central Blue SCC estimate increases that value by 91%. The RICE50+ framework also allows for alternative damage functions. For example, applying the Burke, Hsiang, and Miguel (2015)⁹⁰ damage function, which links temperature directly to GDP growth, yields an SCC of \$253, in which case adding the Blue SCC would imply an 18% increase. On the other end of the spectrum, using the original Nordhaus damage function results in an SCC of only \$21.2, which would imply that incorporating blue damages increases the SCC by approximately 218%. Importantly, to avoid double counting when comparing against SCCs implied by macro-damage functions—which, by construction, capture market channels in GDP—we net out the market-use component of the Blue SCC and add only the non-market welfare component to the baseline SCC for these percentage increases. While we report the central comparison using Kalkuhl and Wenz, we acknowledge that all damage functions remain subject to considerable uncertainty and debate. This reinforces the importance of systematically accounting for blue capital within the SCC framework, regardless of the baseline used.

Substitutability Intuition

Let welfare depend on market consumption M and a non-market benefit N . Take a current bundle $M = 6, N = 4$. Let welfare be given by $U(M, N) = (M^\theta + N^\theta)^{\frac{1}{\theta}}$. If N falls from 4 to 3 (a loss of 1 unit), the market-equivalent compensation is the increase in M that keeps welfare unchanged and can be obtained by dividing the partial derivative of U with respect to N by the partial derivative of U with respect to M . With $\theta = 1$ (i.e. perfect substitutes), a 1-unit loss of N requires +1 in M . For imperfect substitution (e.g. $\theta = -1$), a 1-unit loss of N requires +2.25 in M . Thus, losing 1 unit of a non-market good generally requires more than one market dollar to hold welfare constant when goods are imperfect substitutes. This mechanism underlies the higher values assigned to non-market and non-use categories after welfare adjustment and explains the transformation from raw damages in Fig. 3A to welfare-adjusted damages in Fig. 3C. Supplementary Fig. S26 shows the point-by-point mapping from pre-adjusted to post-adjusted damages.

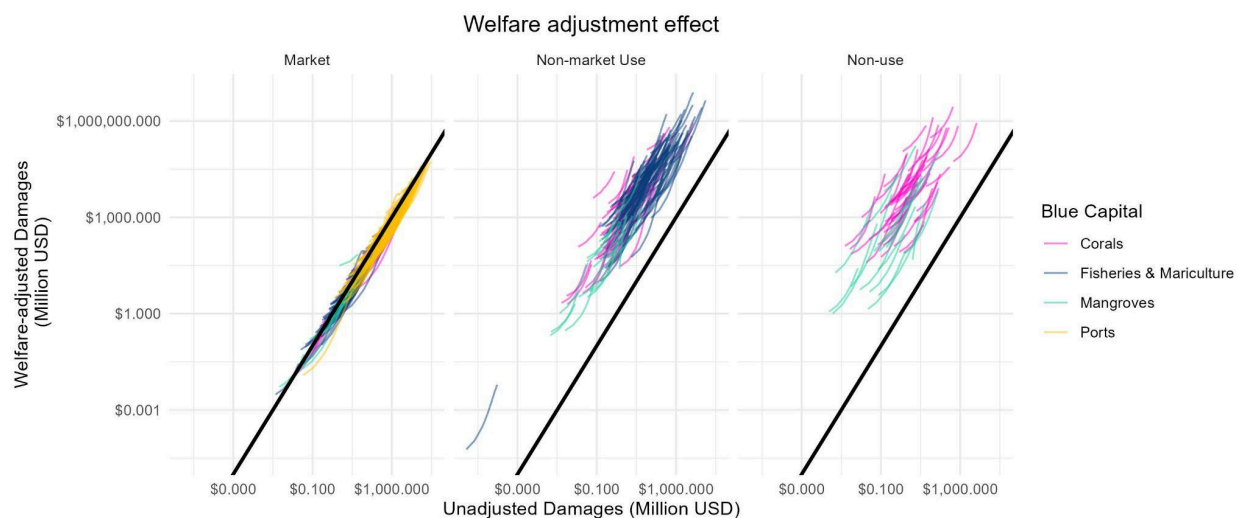


Figure S26. Welfare adjustment effect.

Unadjusted (x-axis) versus welfare-adjusted (y-axis) damages for each country–sector, by value type (Market, Non-market use, Non-use). The 45° identity line indicates no adjustment. Market damages cluster near this line; non-market and non-use damages lie above it, reflecting larger market-equivalent losses when market and non-market benefits are imperfect substitutes. Colors denote sectors (corals, fisheries and mariculture, mangroves, ports). All values are in 2020 USD (billions). Each line traces annual damages for a country–sector from 2020 to 2100 (typically moving from lower left to upper right as damages rise over the century).

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